ROMA TRE
UNIVERSITÀ DEGLI STUDI

INGEGNERIA
DIPARTIMENTO DI ECCELLENZA

Information Visualization

Orthogonal Drawings by Incremental Approaches

M. Patrignani

 copyright©2024 m.patrignani

Copyright notice

- All the pages/slides in this presentation, including but not limited to, images, photos, animations, videos, sounds, music, and text (hereby referred to as “material”) are protected by copyright
- This material, with the exception of some multimedia elements licensed by other organizations, is property of the authors and/or organizations appearing in the first slide
- This material, or its parts, can be reproduced and used for didactical purposes within universities and schools, provided that this happens for non-profit purposes
- Any other use is prohibited, unless explicitly authorized by the authors on the basis of an explicit agreement
- This copyright notice must always be redistributed together with the material, or its portions

Incremental orthogonal drawings

- Biedl-Kant algorithm
- No-change scenario
- Relative-coordinate scenario

 copyright©2024 m.patrignani

Biedl-Kant algorithm

- Proposed by [Biedl & Kant, 94]
- Assumes that
  - the graph is biconnected
  - each vertex has maximum degree four

 copyright©2024 m.patrignani

Topological sort of st-graphs

- Let G be an st-graph
  - single source s
  - single sink t
- Any topological sort of G has the property that for every v_j, with $0 < j < n - 1$, has at least one "predecessor" v_i and at least one "successor" v_k that are neighbors of v_j, with $i < j < k$

 copyright©2024 m.patrignani

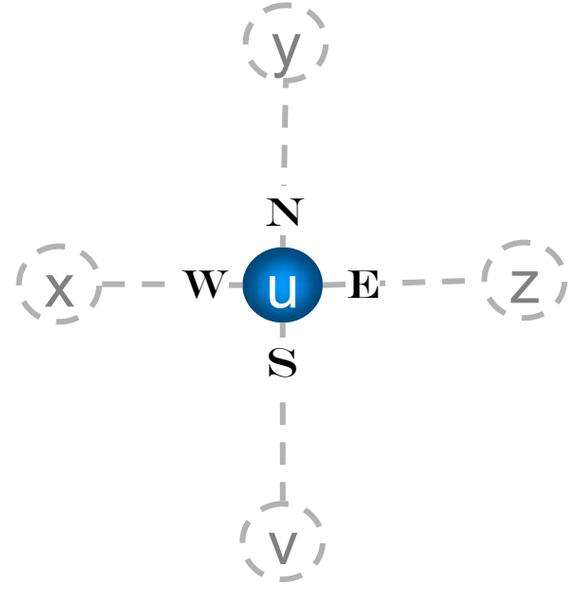
Incremental approach

- Orient the edges of the biconnected graph G so to produce an st-graph
- Consider the nodes of G in the order given by a topological sort of the st-graph
- Start from a fixed drawing of the first two vertices
- Add current node to the drawing

 copyright©2024 m.patrignani

Node ports

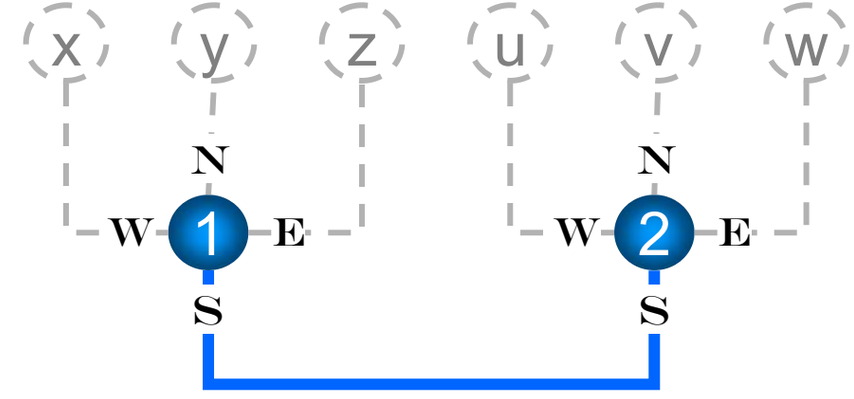
- Each node u is incident to at most four edges
  - they use its four ports N, E, S, W



 copyright©2024 m.patrignani

Incremental approach

- The initial drawing has only the first two nodes connected with 2 bends via their South port
- Each node is added to the drawing using a new row on the top of it



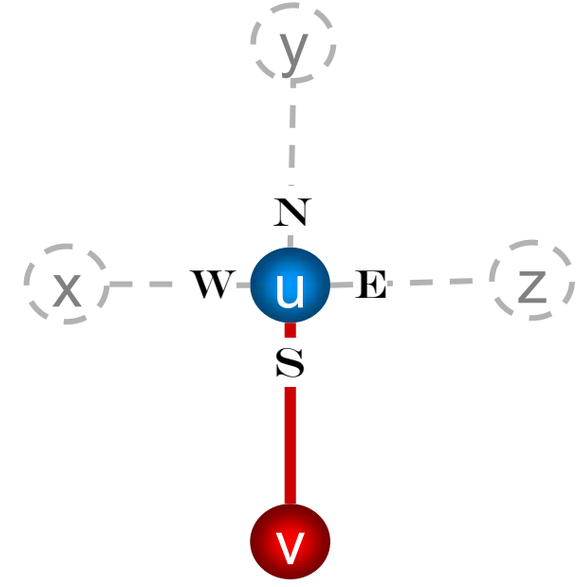
New row for new node

Current drawing

copyright©2024 m.patrignani

Node ports

- When a node is added to the drawing its South port is surely used
  - the node is attached with a straigth edge to a previously drawn node

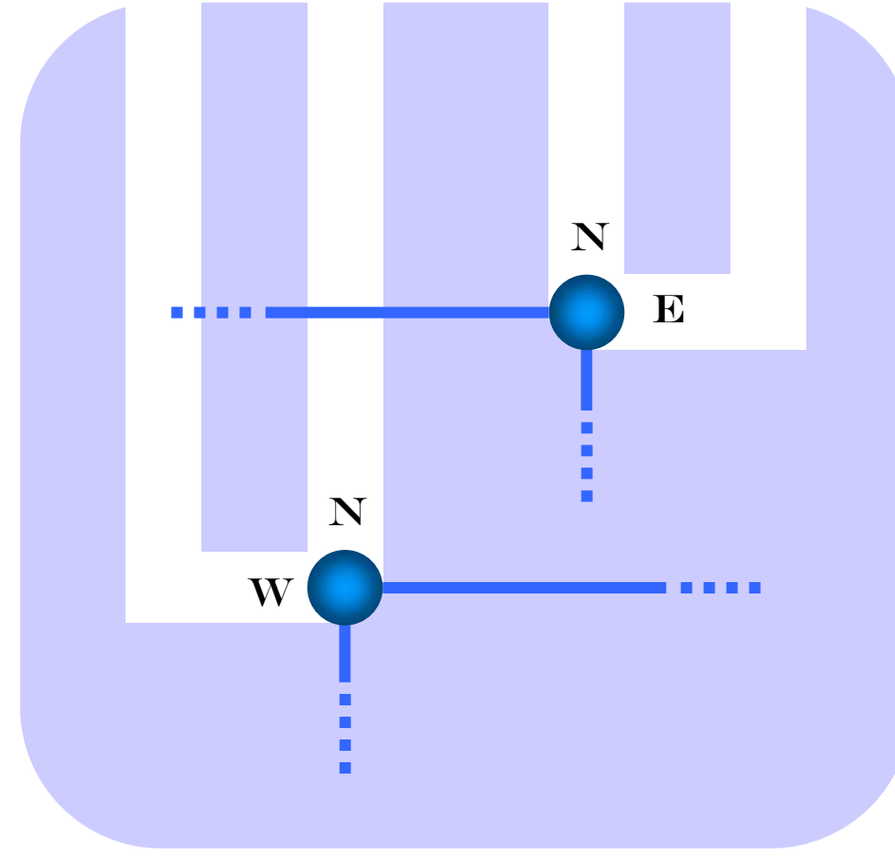


Previous node

 copyright©2024 m.patrignani

Drawing invariant

- These invariants will be preserved throughout the drawing process
  - South ports are always used
  - other unused ports are always reachable from top with at most one bend

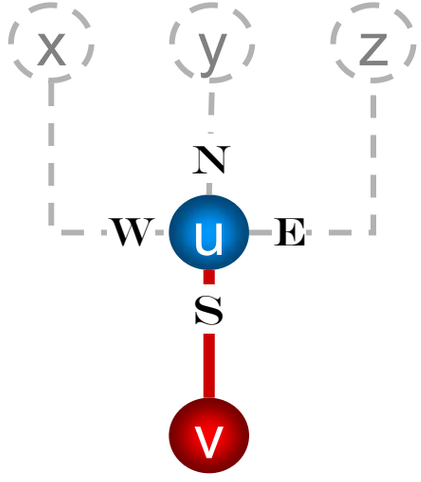


Bounding box of current drawing

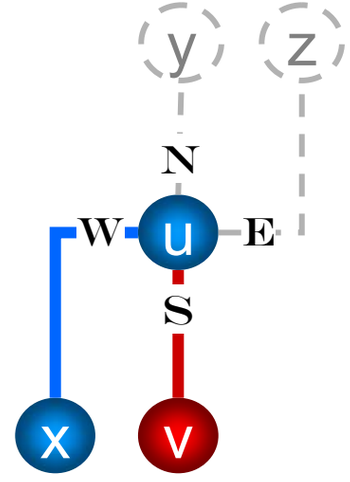
 copyright©2024 m.patrignani

Node ports

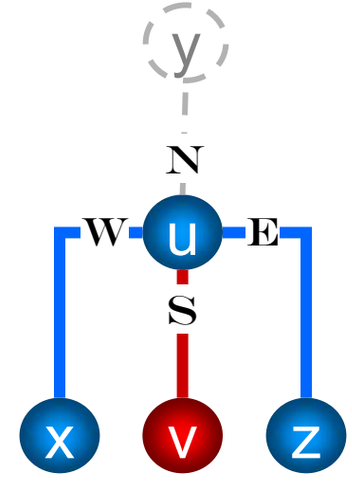
- Depending on the number of connections with previously drawn nodes several configurations are possible
  - node v is chosen in such a way to avoid the leftomost or rightmost of previously drawn nodes if possible



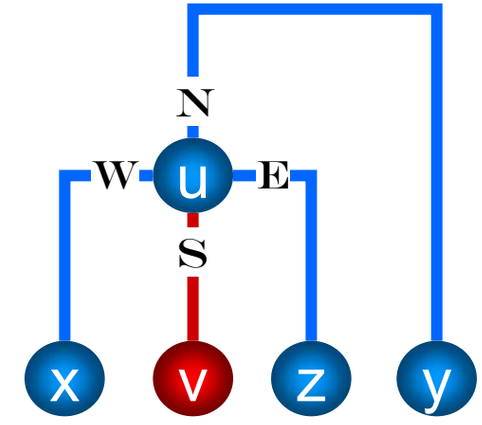
One connection



Two connections



Three connections

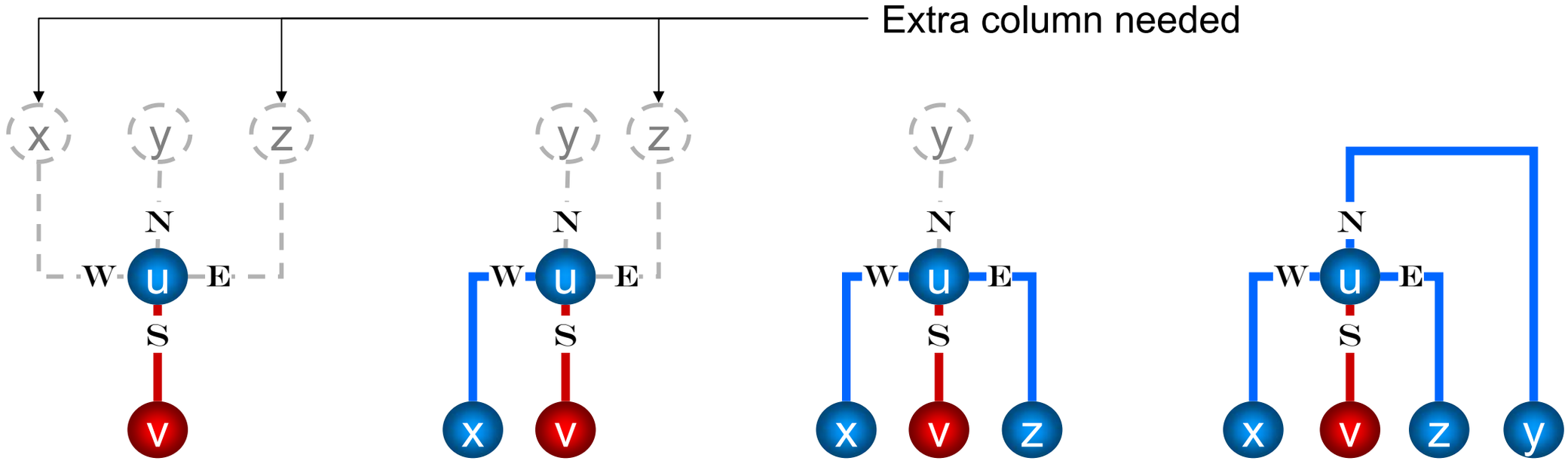


Four connections

copyright©2024 m.patrignani

Extra columns

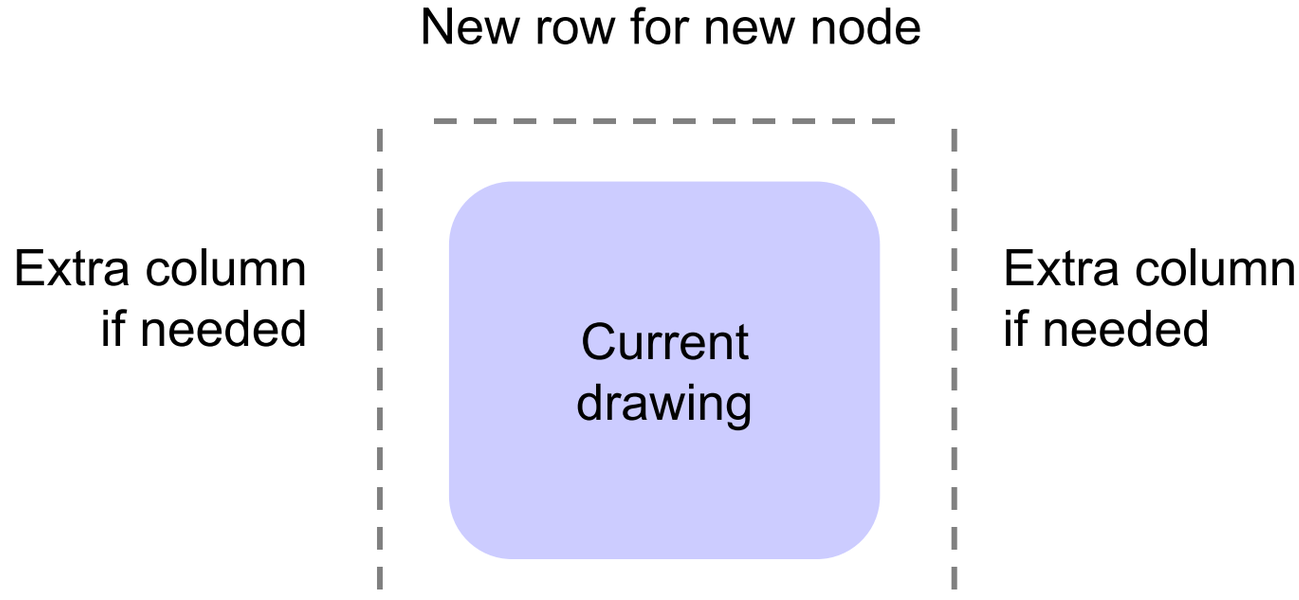
- Extra columns are needed for each connection to nodes that will be inserted and that do not use the North port



copyright©2024 m.patrignani

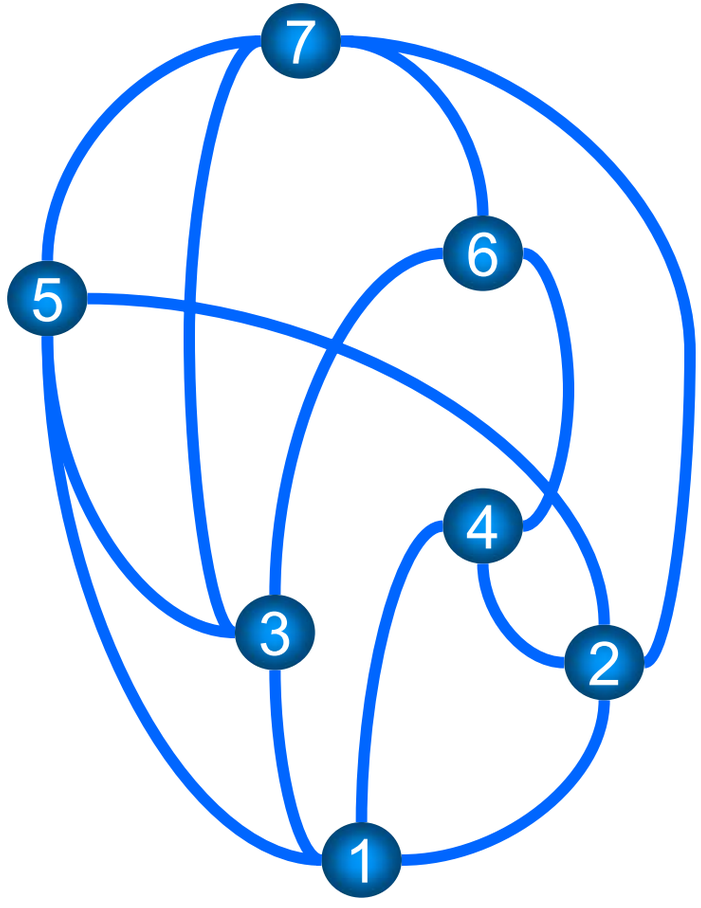
Allocation of extra columns

- Extra columns are allocated on the left and right of the current drawing when they are used



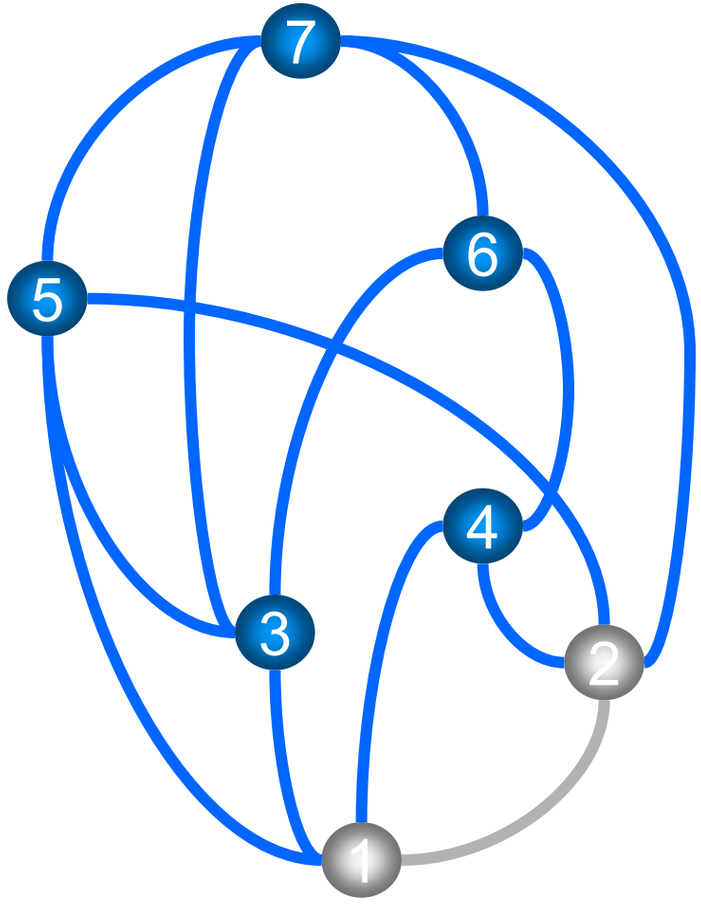
 copyright©2024 m.patrignani

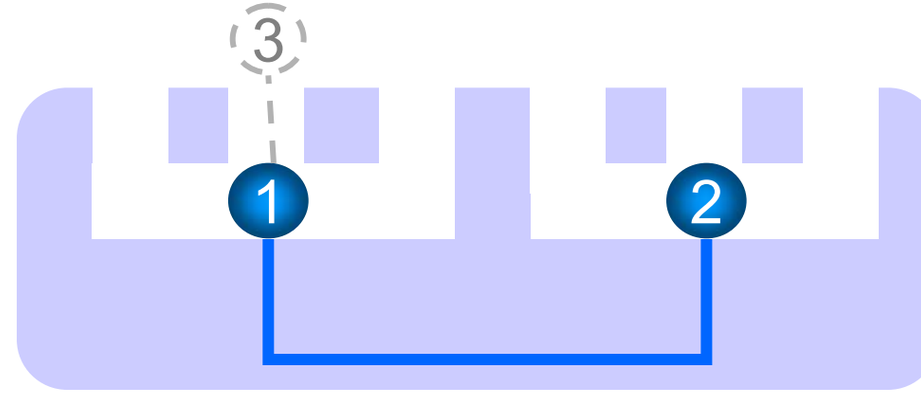
Example



 copyright©2024 m.patrignani

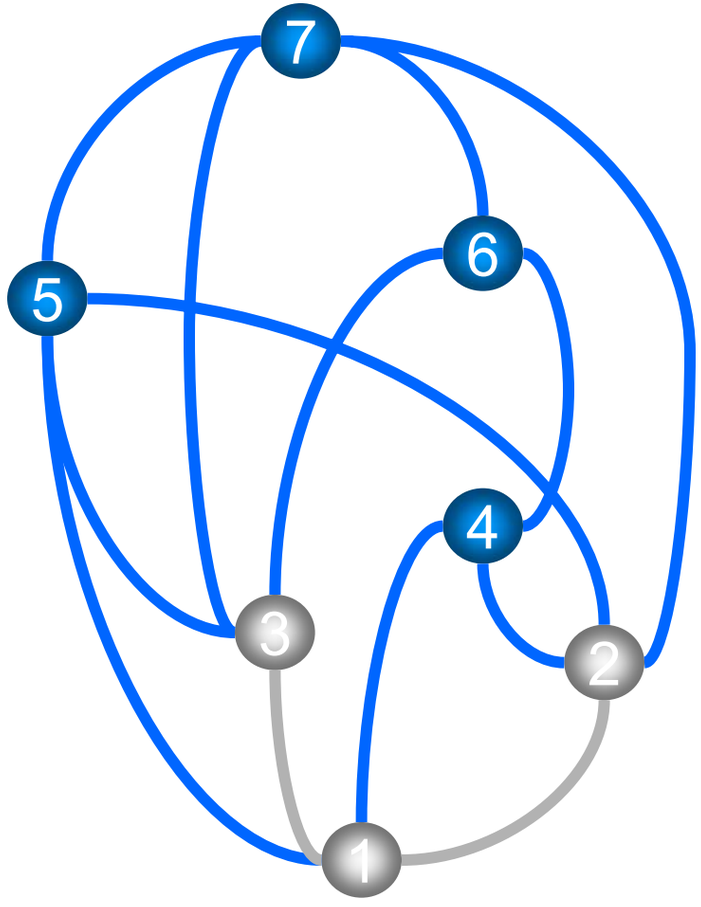
Example

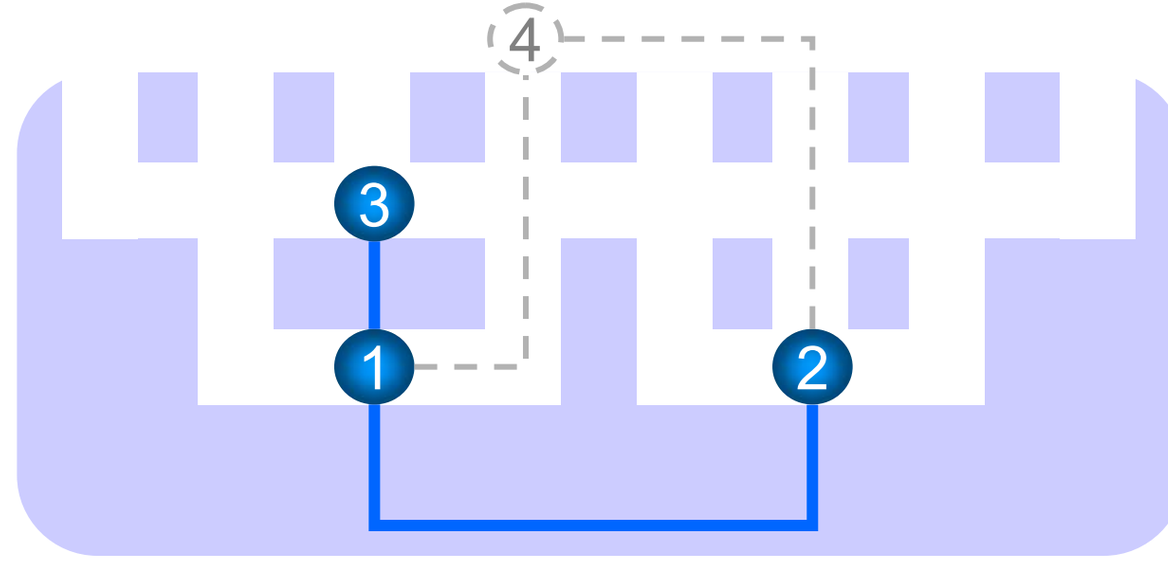




 copyright©2024 m.patrignani

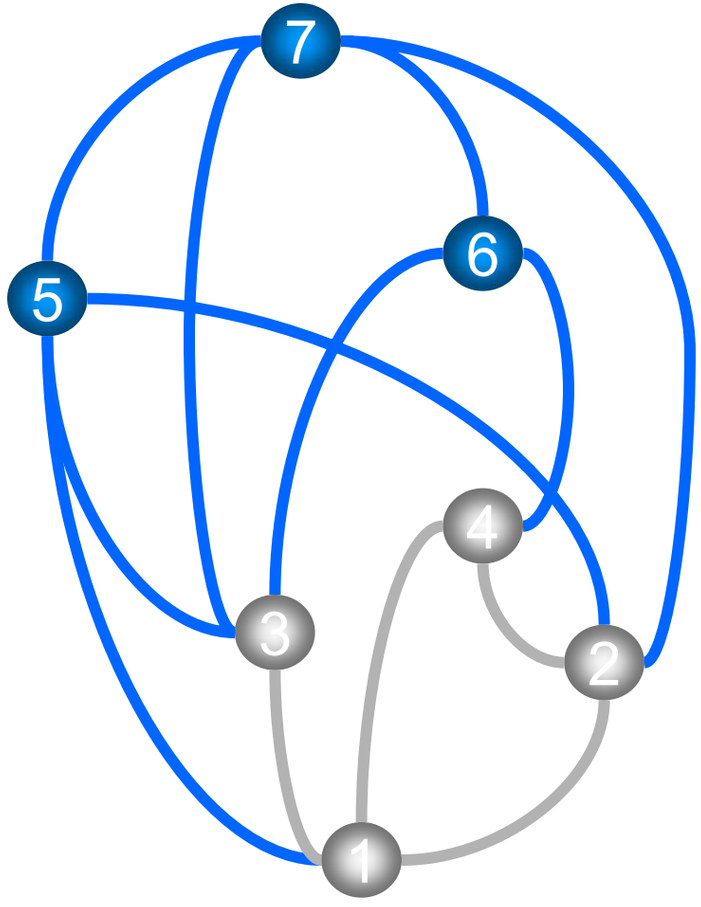
Example

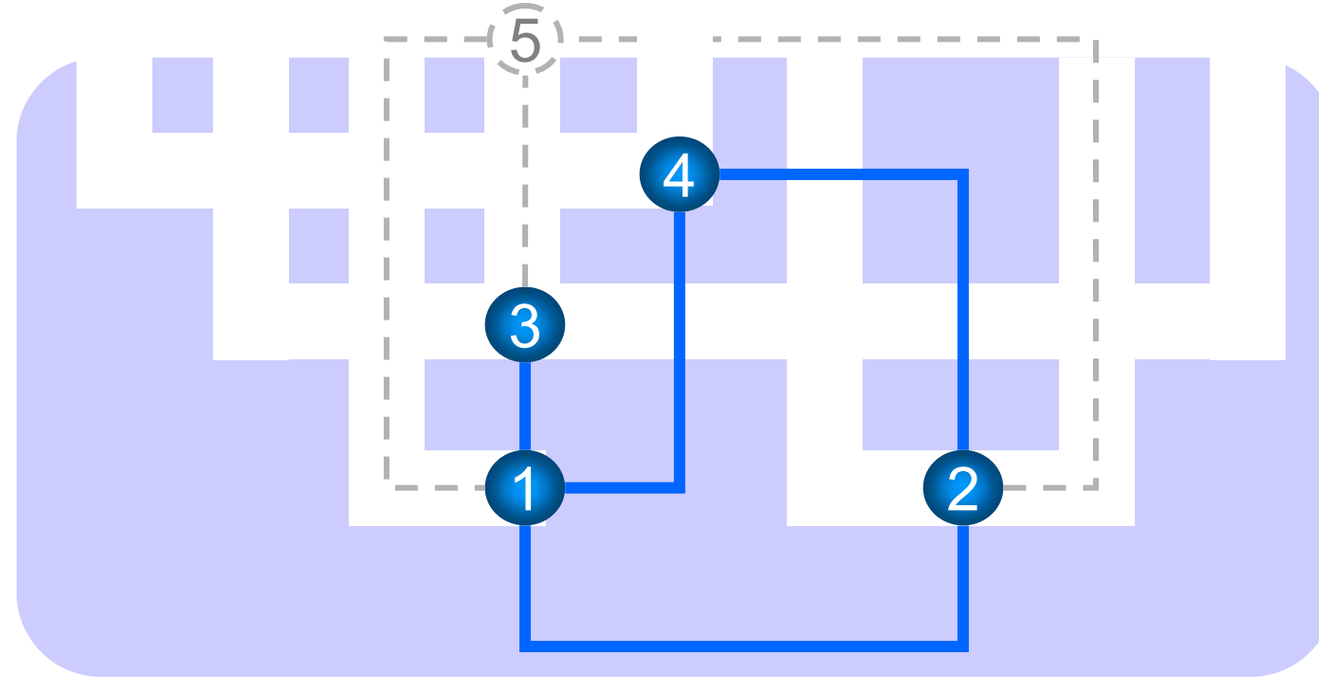




 copyright©2024 m.patrignani

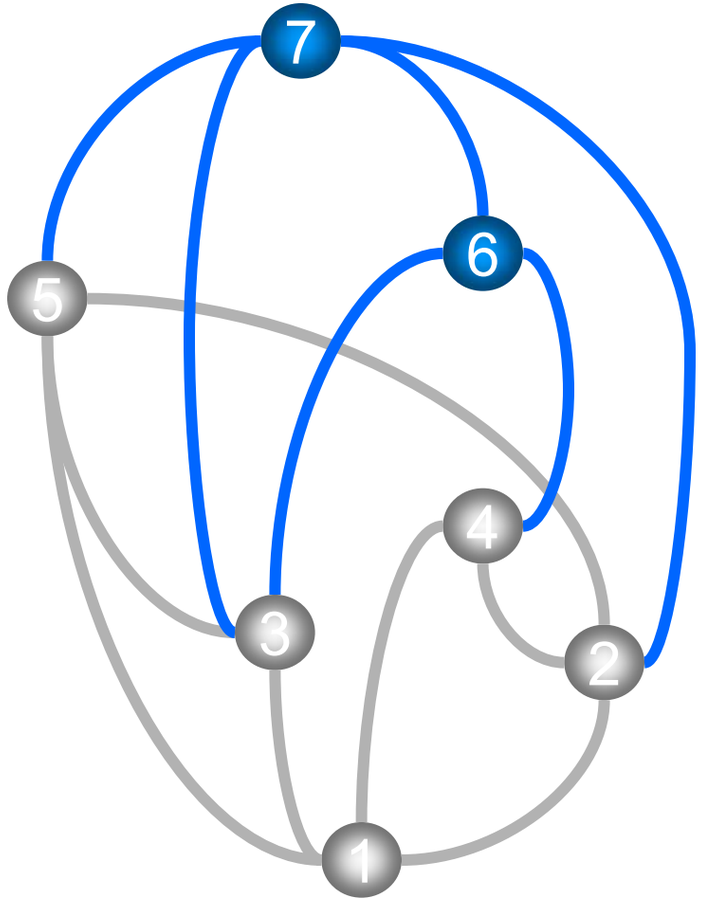
Example

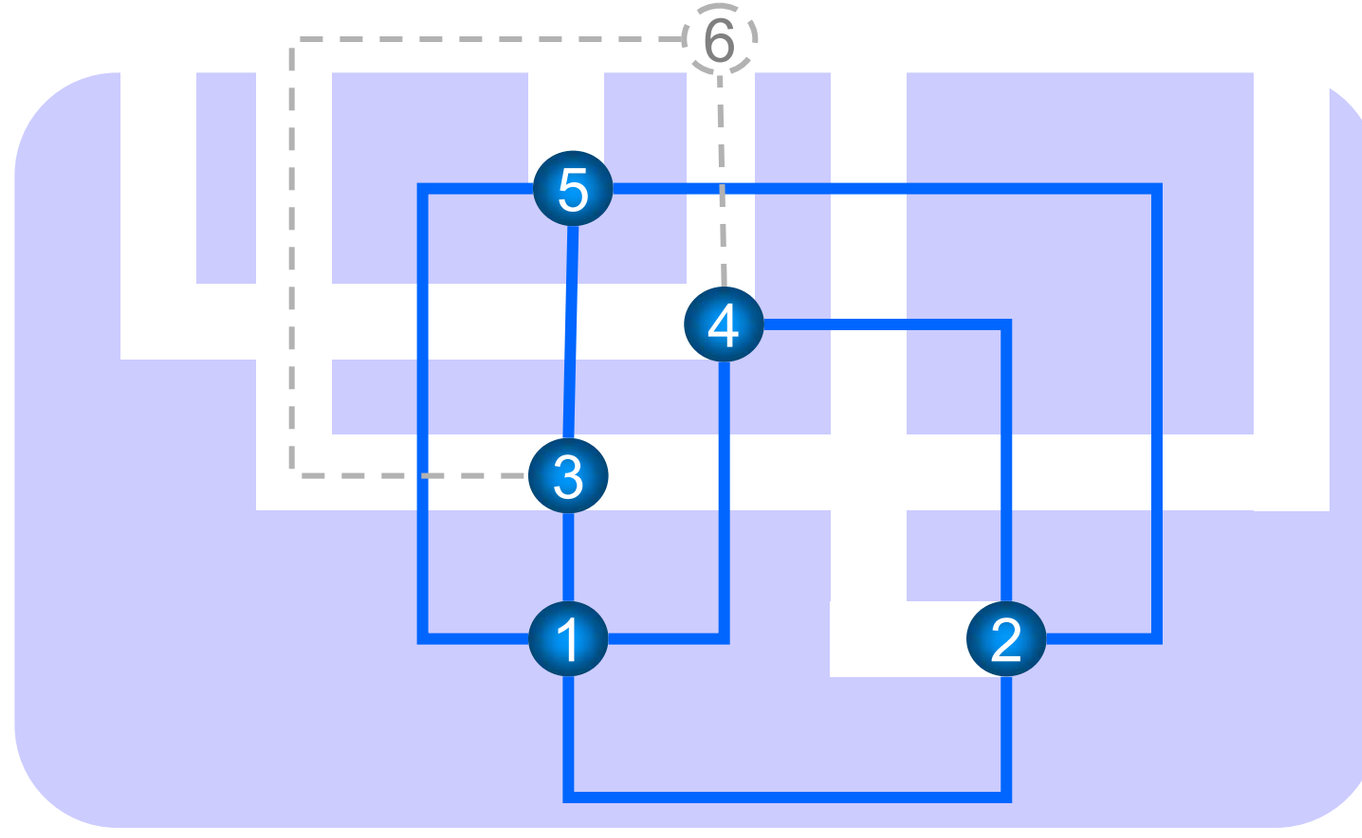




 copyright©2024 m.patrignani

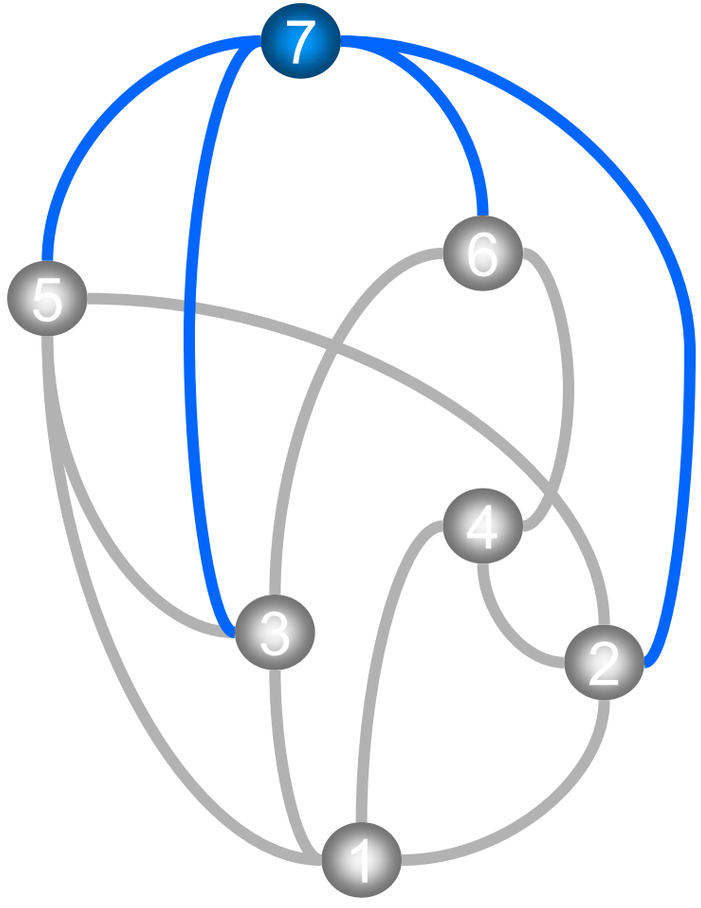
Example

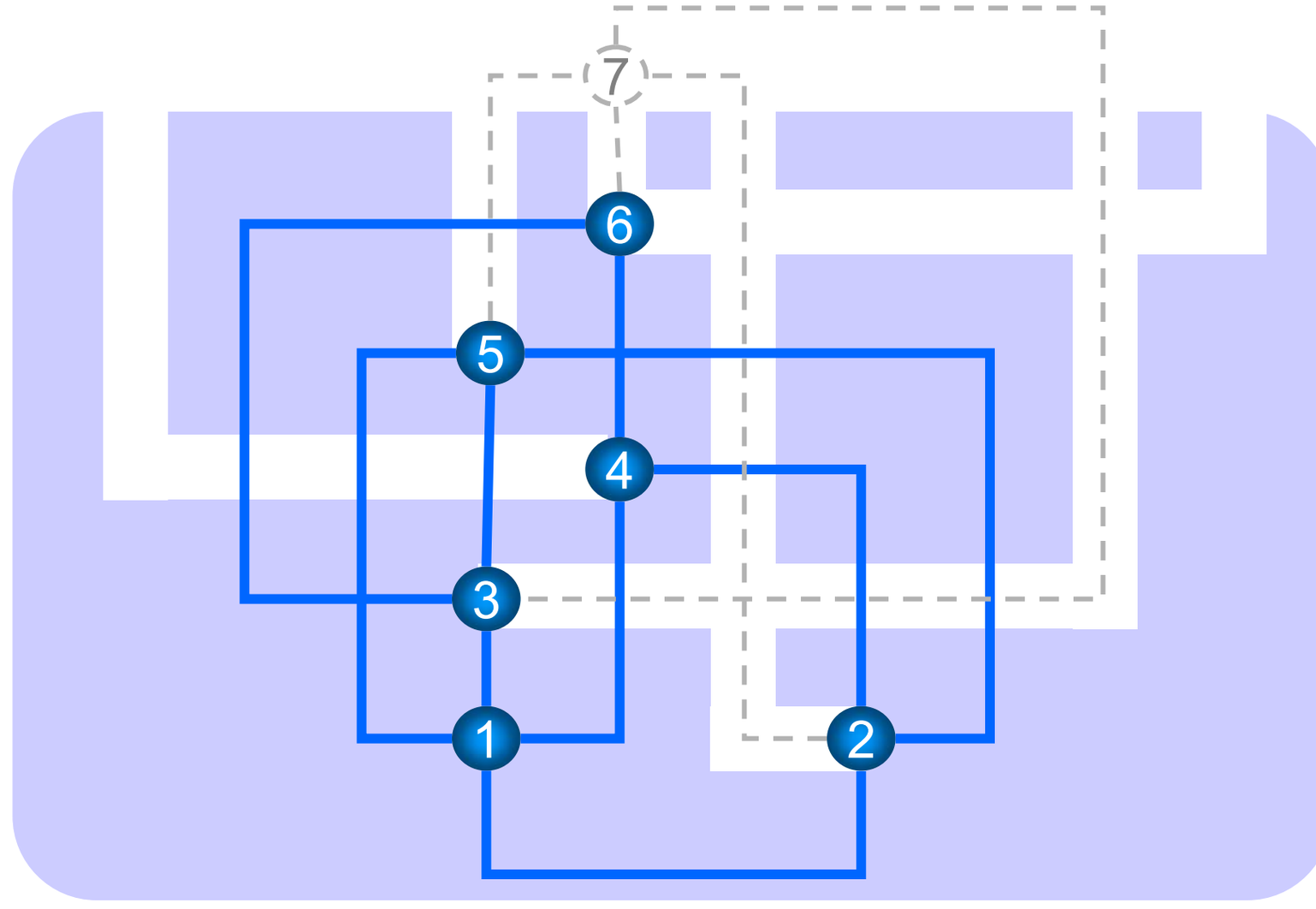




 copyright©2024 m.patrignani

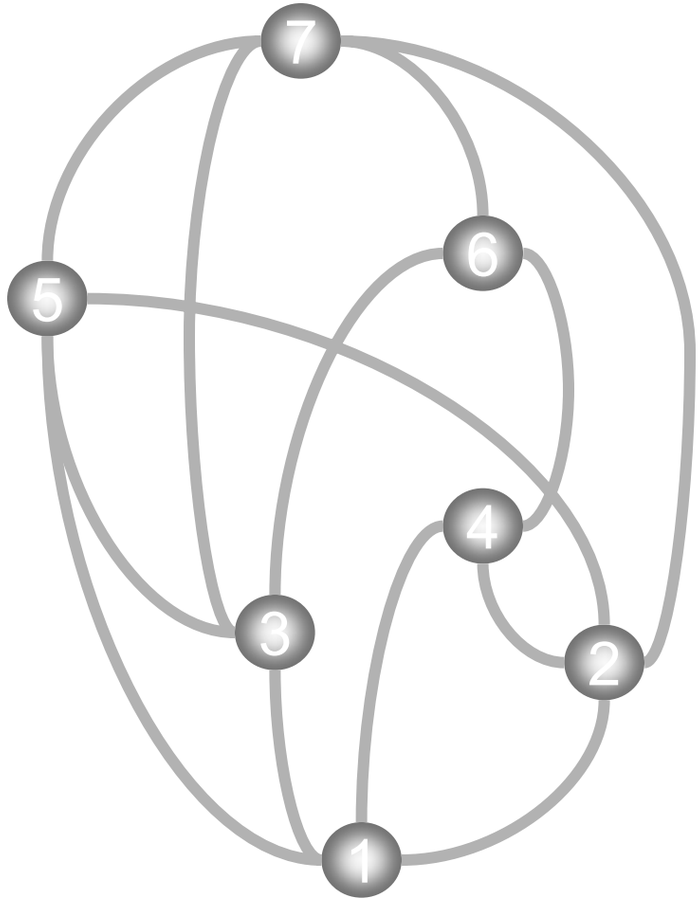
Example

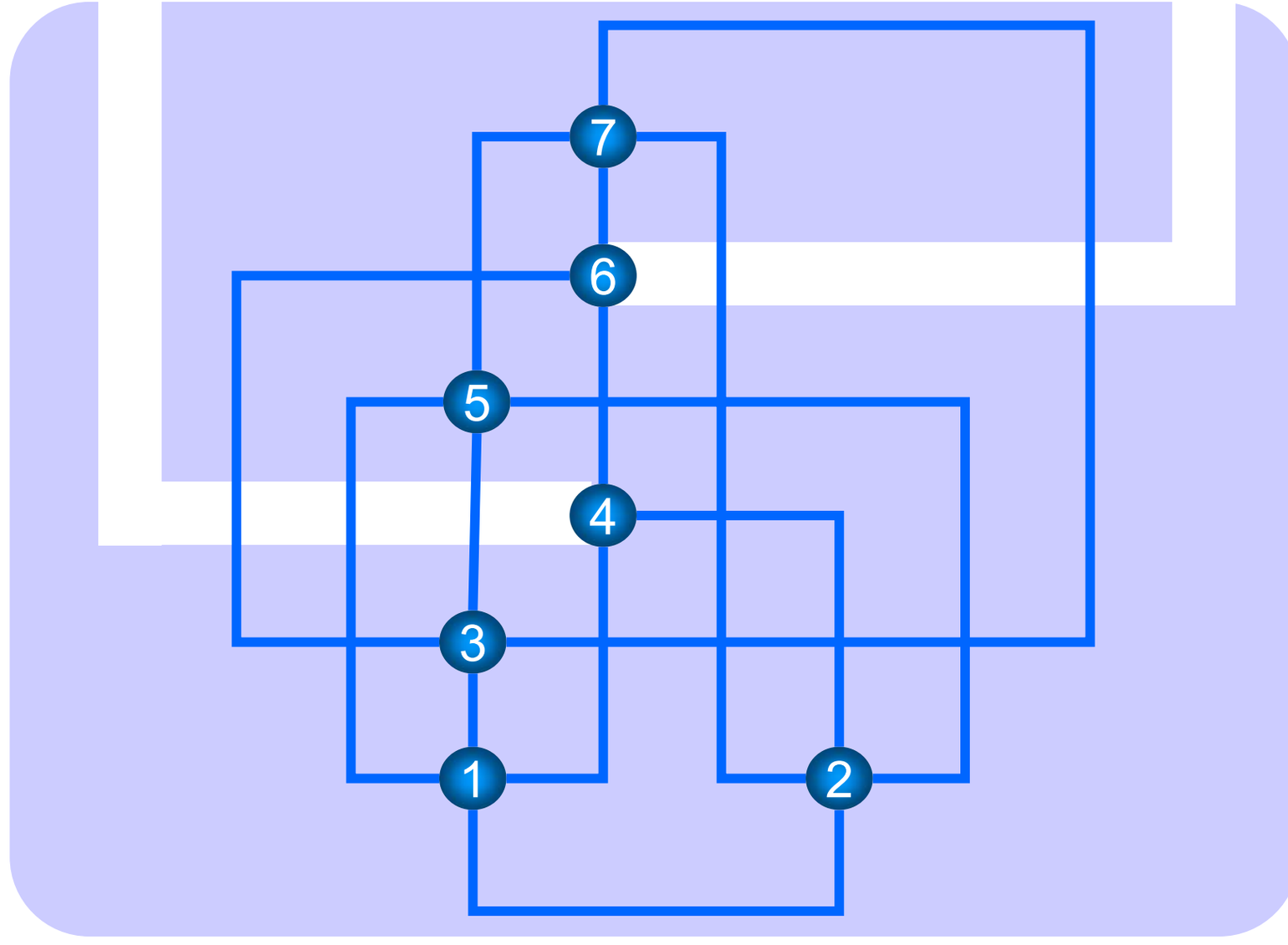




 copyright©2024 m.patrignani

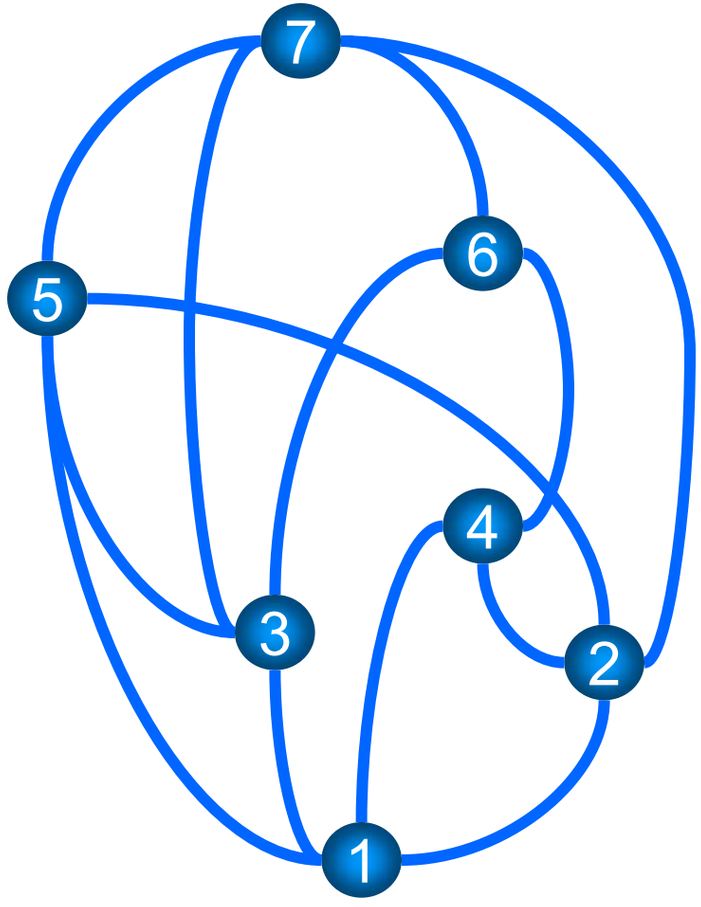
Example

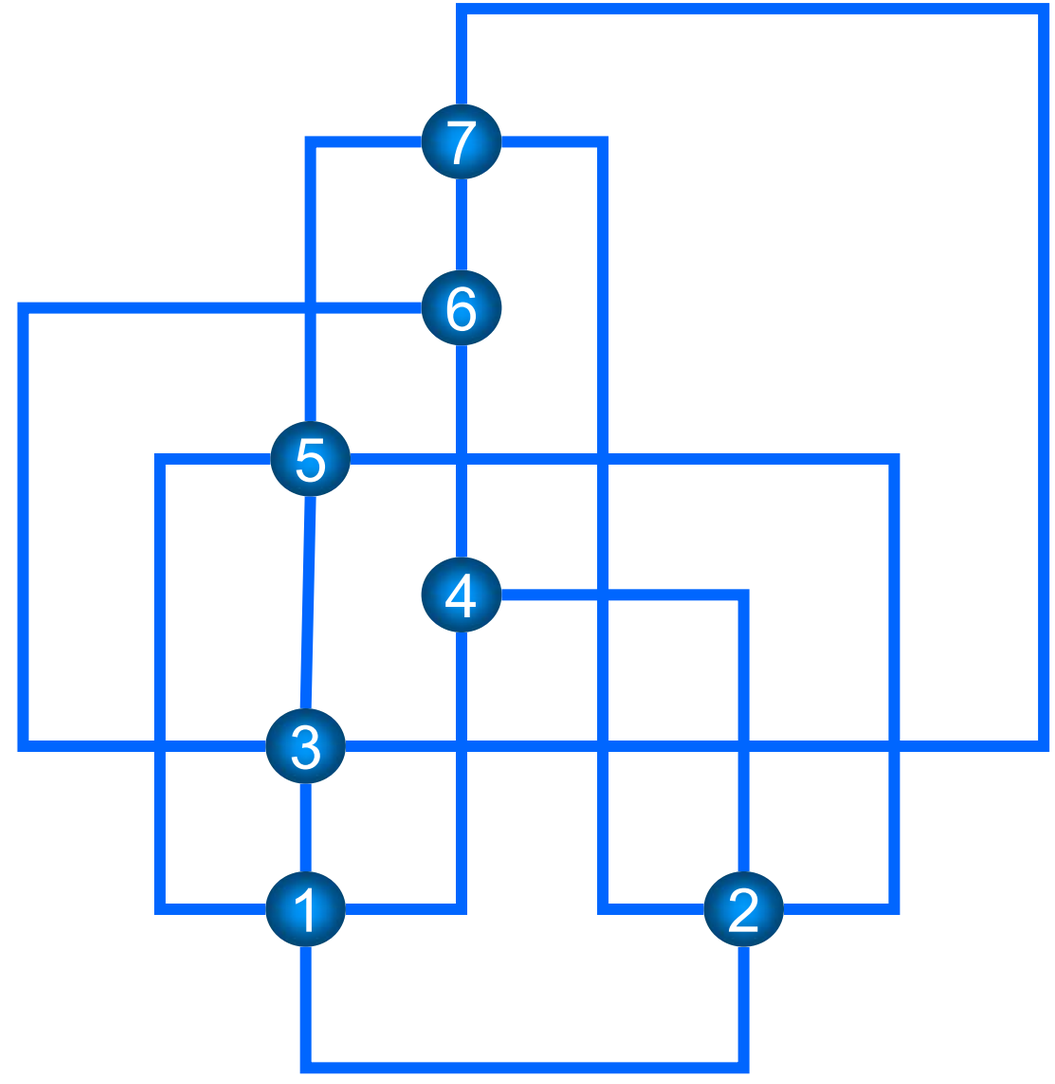




 copyright©2024 m.patrignani

Example





 copyright©2024 m.patrignani

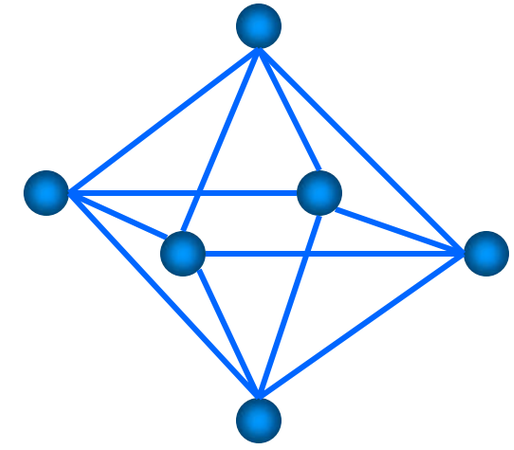
Properties of the Biedl-Kant drawings

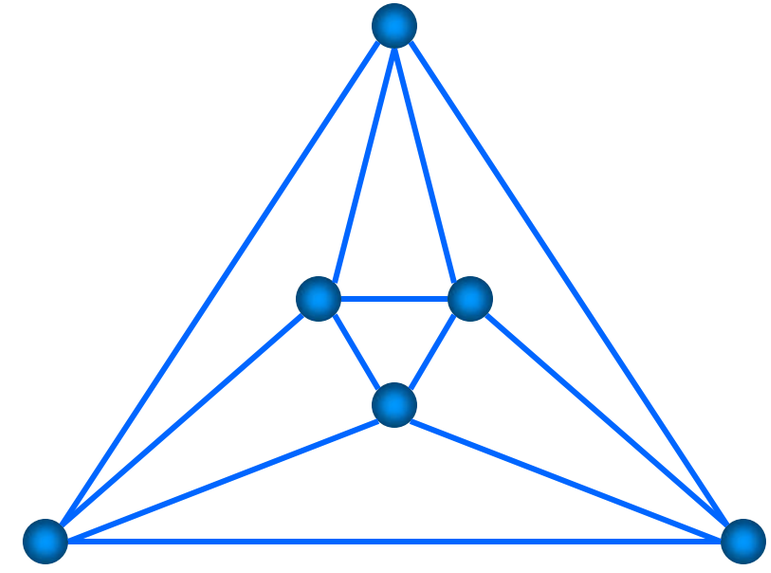
- It can be proved that
  - the grid size is at most $(m - n + 1) \times n$
  - at most one edge has three bends, all other edges have at most two bends
  - overall, there are at most $2m - 2n + 4$ bends in the drawing

 copyright©2024 m.patrignani

Refinements of the algorithm

- In [Biedl & Kant, 94] it is shown that
  - edges with three bends can be avoided unless G is the octahedron
    - i.e., the unique planar 4-regular graph with six vertices
  - by using a special topological sort the following bounds can be obtained
    - the grid size is n × n
    - at most 2n + 2 bends
  - further if G is not 4-regular
    - the grid size is (n − 1) × (n − 1)
    - at most 2n − 1 bends





octahedron

copyright©2024 m.patrignani

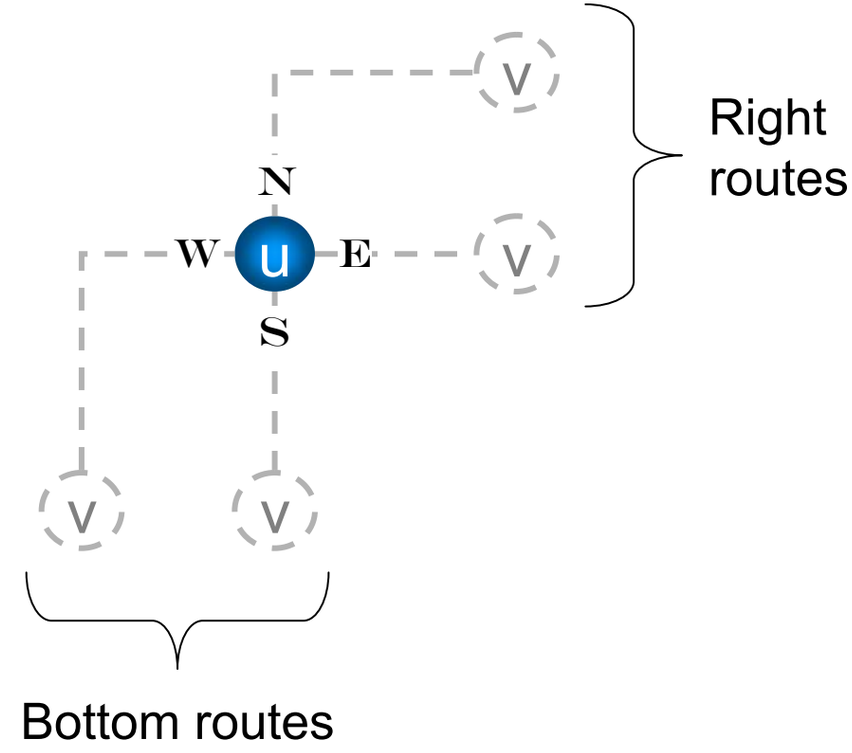
No-Change scenario

- Described in [Papakostas *et al.*, 97] and [Papakostas & Tollis, 98]
- As for the Biedl-Kant algorithm, the drawing system never changes the positions of vertices and bends of the current drawing
  - new node *v* is placed outside the bounding box of existing drawing minimizing rows, columns and bends
- Incremental construction at time t
  - it is only required that added nodes have at least one connection to already drawn nodes

 copyright©2024 m.patrignani

Node routes

- Each node u is incident to at most four edges
  - they use its four ports N, E, S, W
- Each node is reachable in one of four different ways, called “routes”

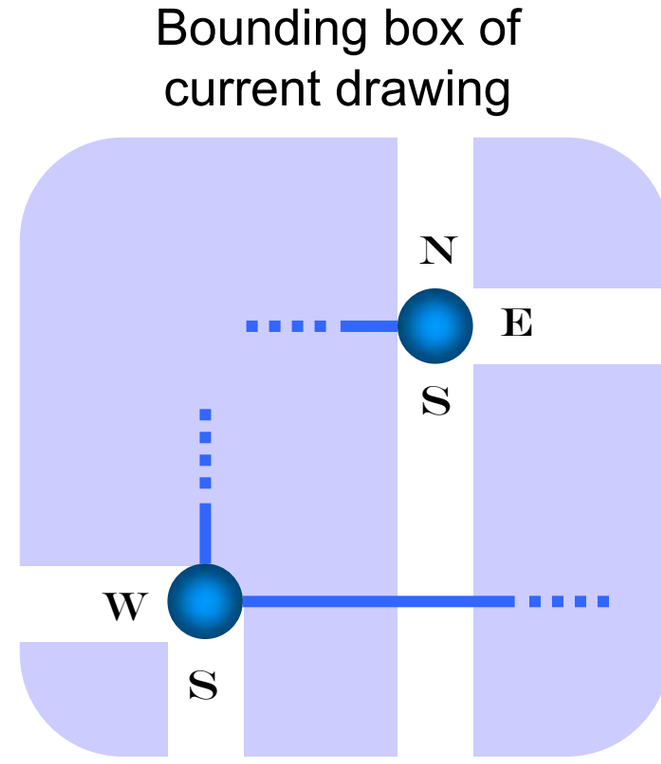


  - the two routes that attach to N and E are called *right routes*
  - the two routes that attach to S and W are called *bottom routes*

 copyright©2024 m.patrignani

Drawing invariant

- This invariant will be preserved throughout the drawing process
  - if a node *n* has an unused port among N, E, S, W, then the halfline starting from *n* and going towards N, E, S, W does not host nodes nor bends

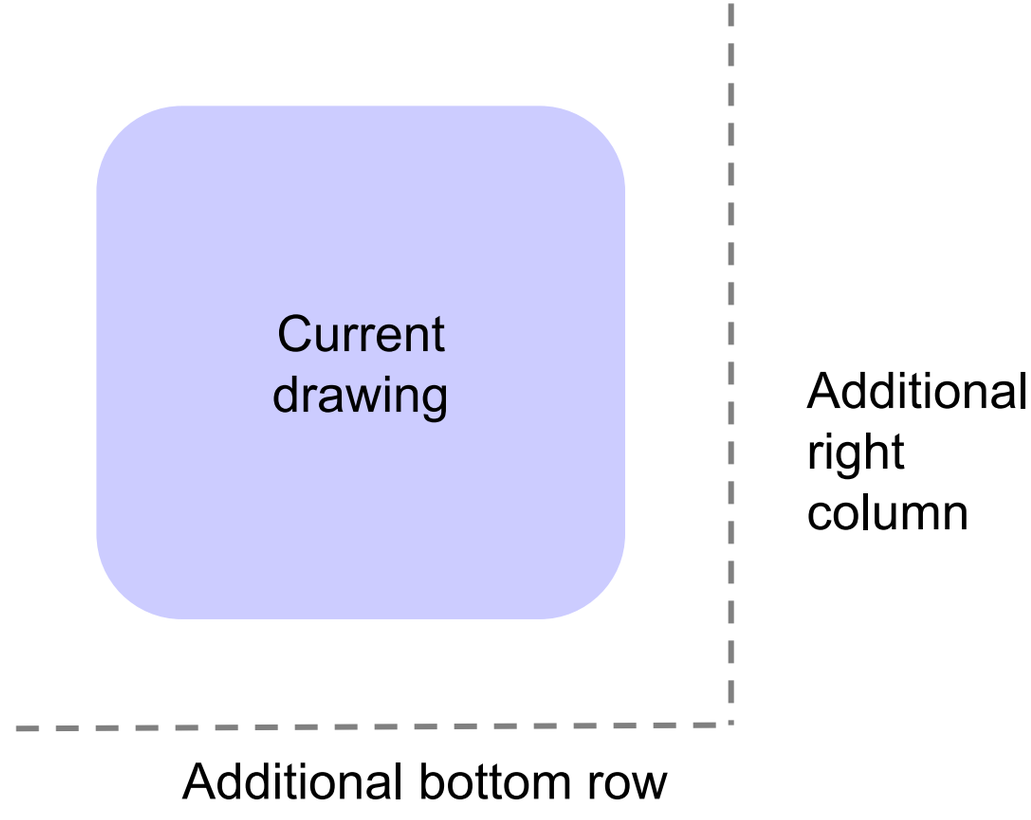


- we say that the unused ports of *n* are “reachable” from ouside

 copyright©2024 m.patrignani

New node placement

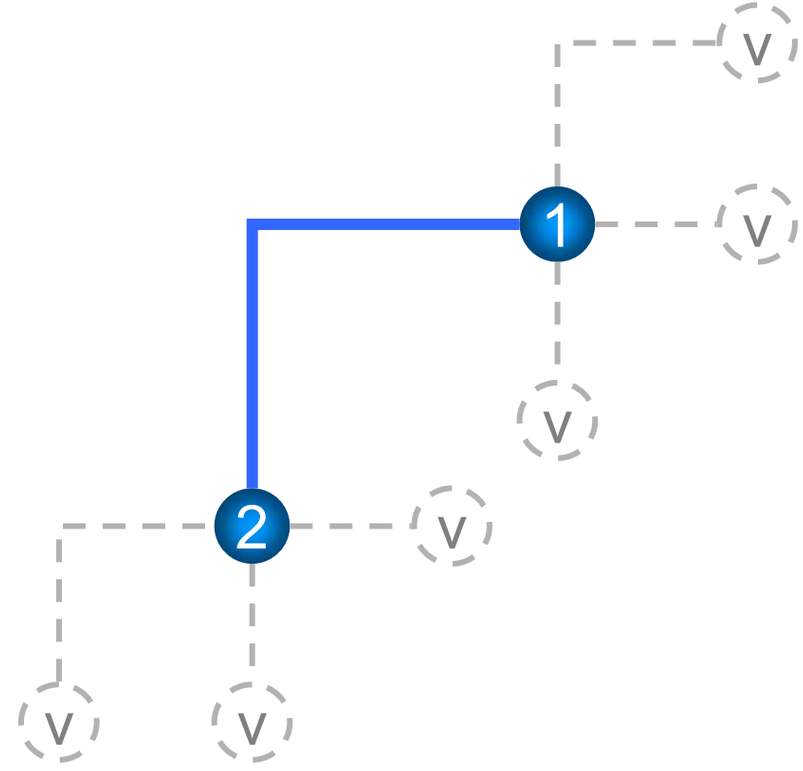
- New nodes will be placed on an additional right column or bottom row



 copyright©2024 m.patrignani

Start drawing

- The first two nodes have a fixed drawing
  - observe that they meet the invariant



 copyright©2024 m.patrignani

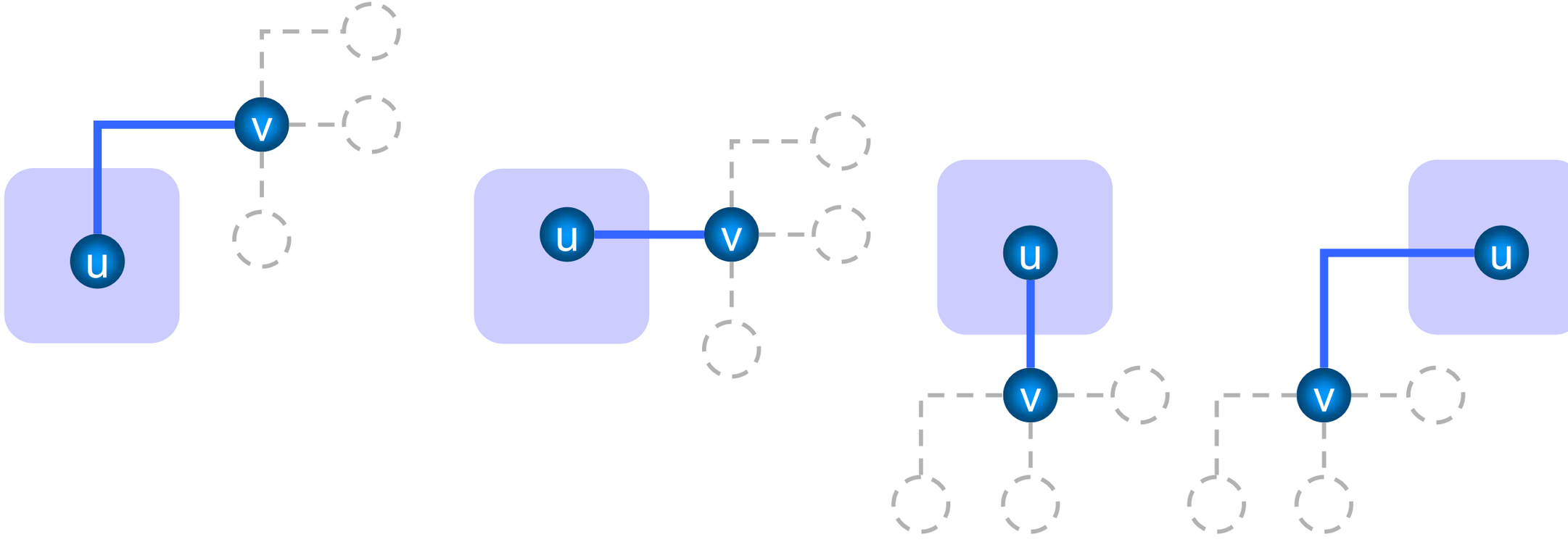
Adding nodes to the drawing

- Since the graph is always connected, a node v to be inserted has local degree one, two, three, or four
- The four cases are considered separately

 copyright©2024 m.patrignani

Adding a degree-one node

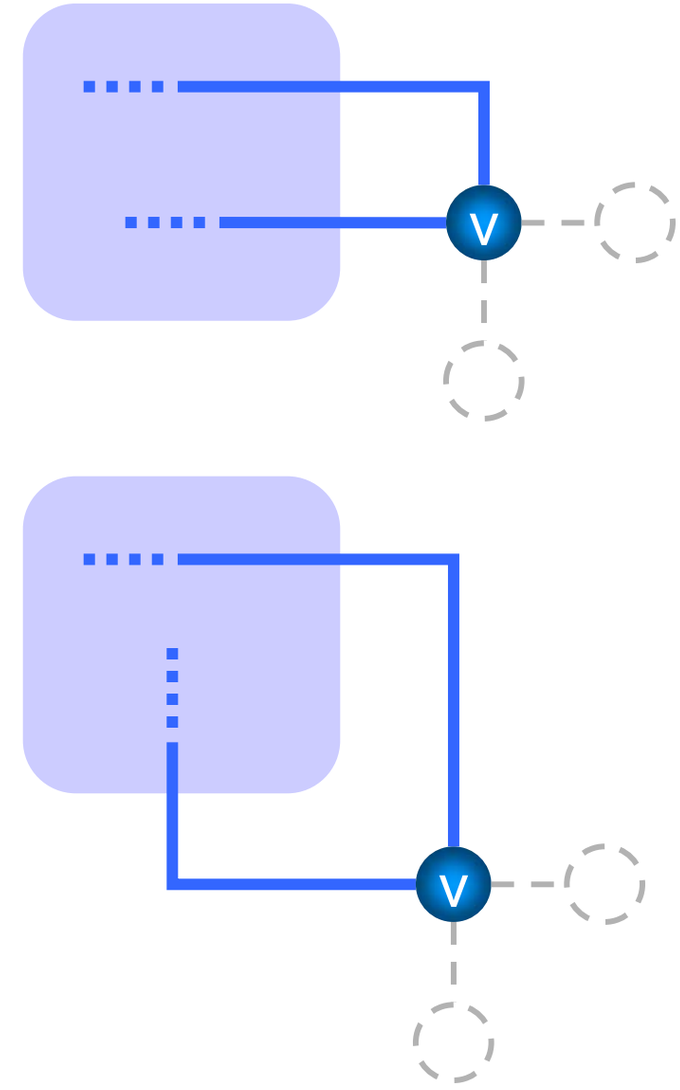
- A degree one vertex can be attached in four different ways depending on the side of node u it attaches to



 copyright©2024 m.patrignani

Adding a degree-two node

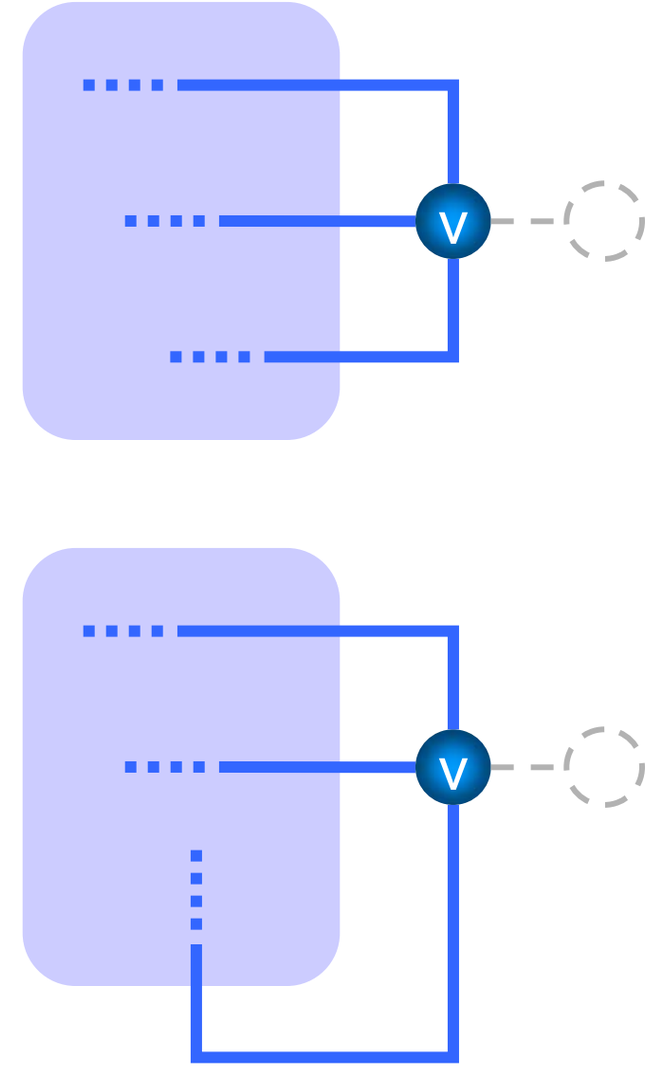
- Requires in the worst case
  - a new row and a new column
  - four bends
    - maximum two on the same edge



 copyright©2024 m.patrignani

Adding a degree-three node

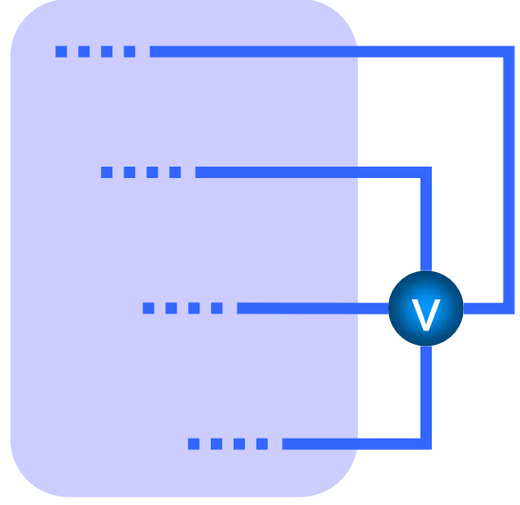
- Requires in the worst case
  - a new row and a new column
  - six bends
    - maximum three on the same edge

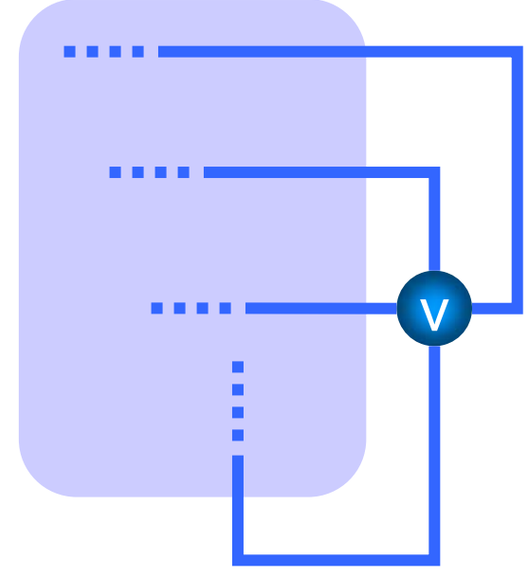


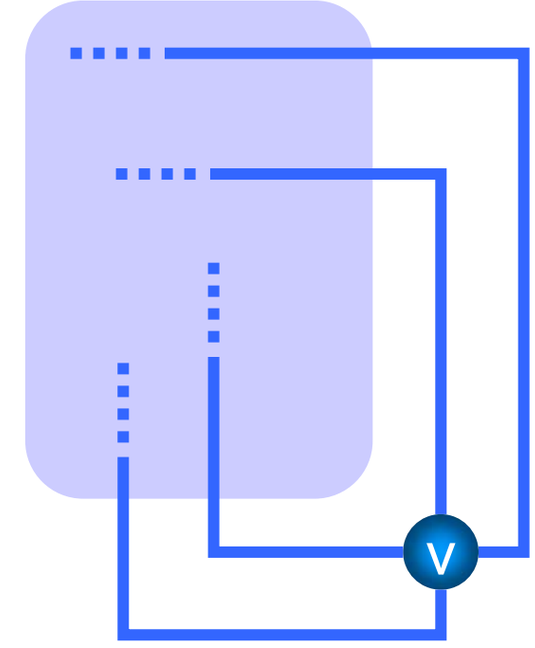
 copyright©2024 m.patrignani

Adding a degree-four node

- Requires in the worst case
  - two new rows and two new columns
  - ten bends
    - maximum three on the same edge







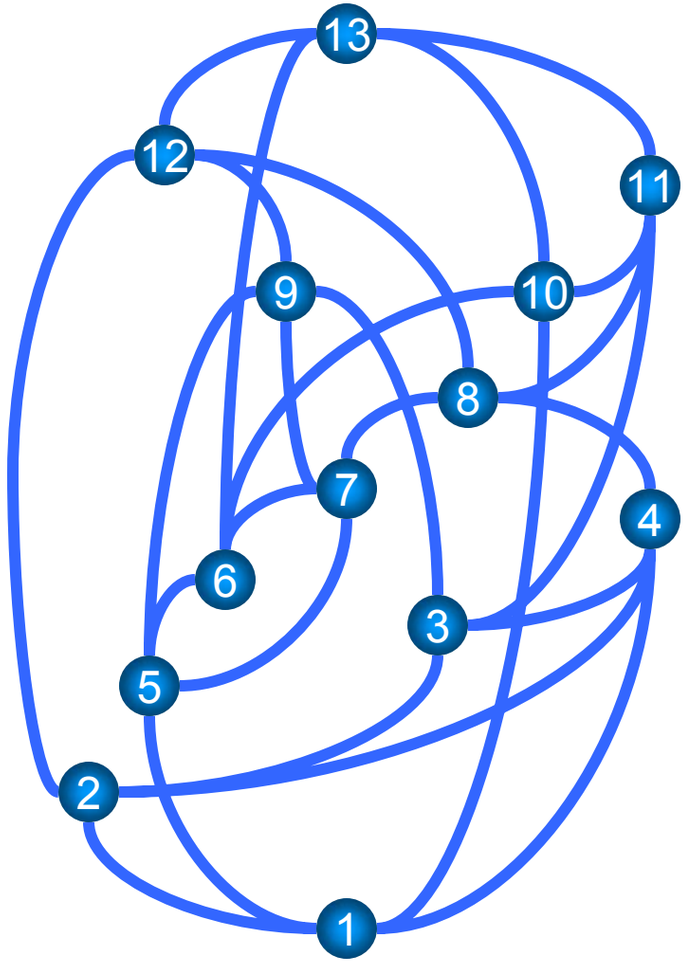
 copyright©2024 m.patrignani

No-Change performances

- It can be proved that the No-Change drawing method produces orthogonal drawings with
  - at most 3 bends per edge
  - at most 8n/3+2 total bends
  - at most $(4n/3)^2$ area

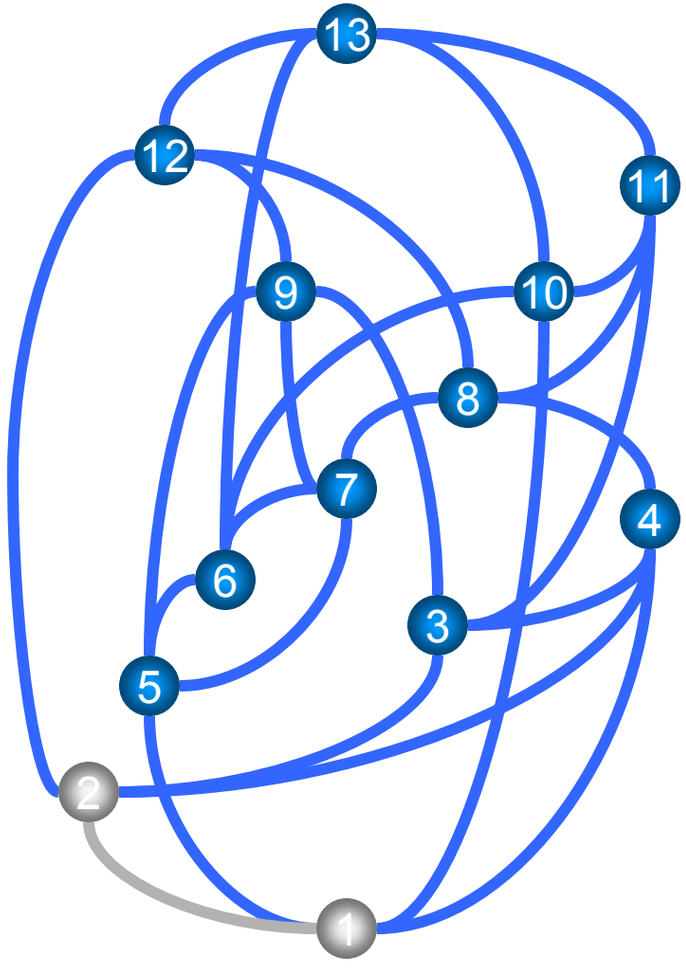
 copyright©2024 m.patrignani

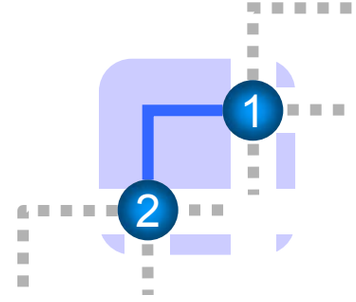
Example



 copyright©2024 m.patrignani

Example

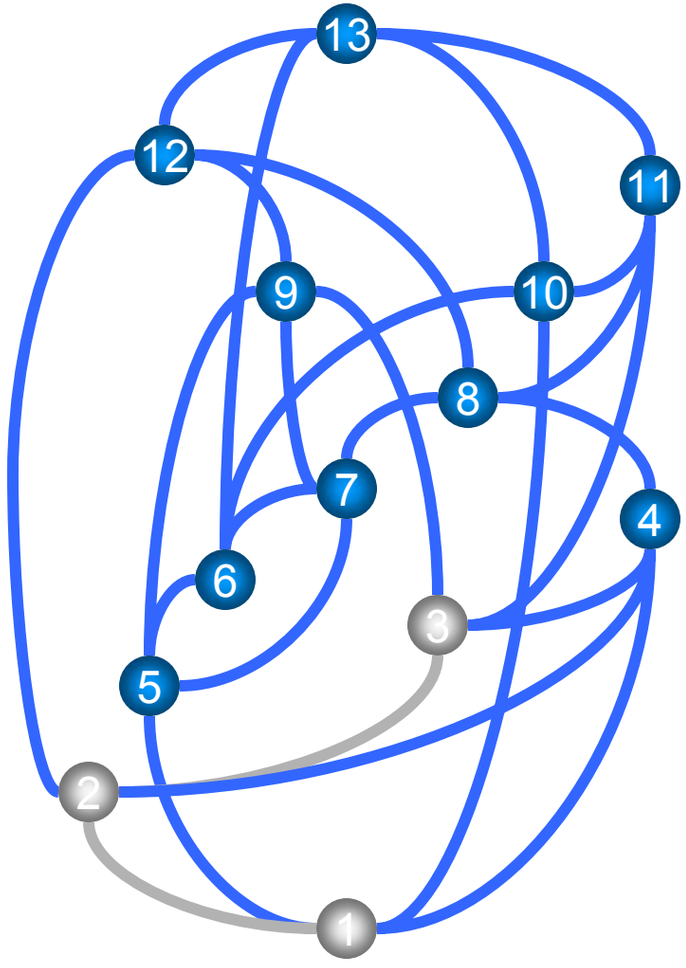


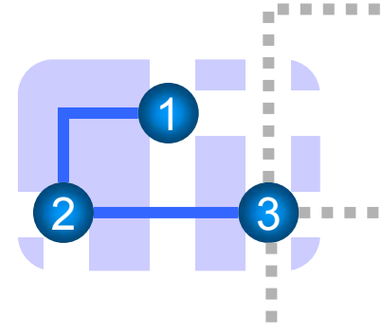


- The first two nodes have a fixed drawing

 copyright©2024 m.patrignani

Example

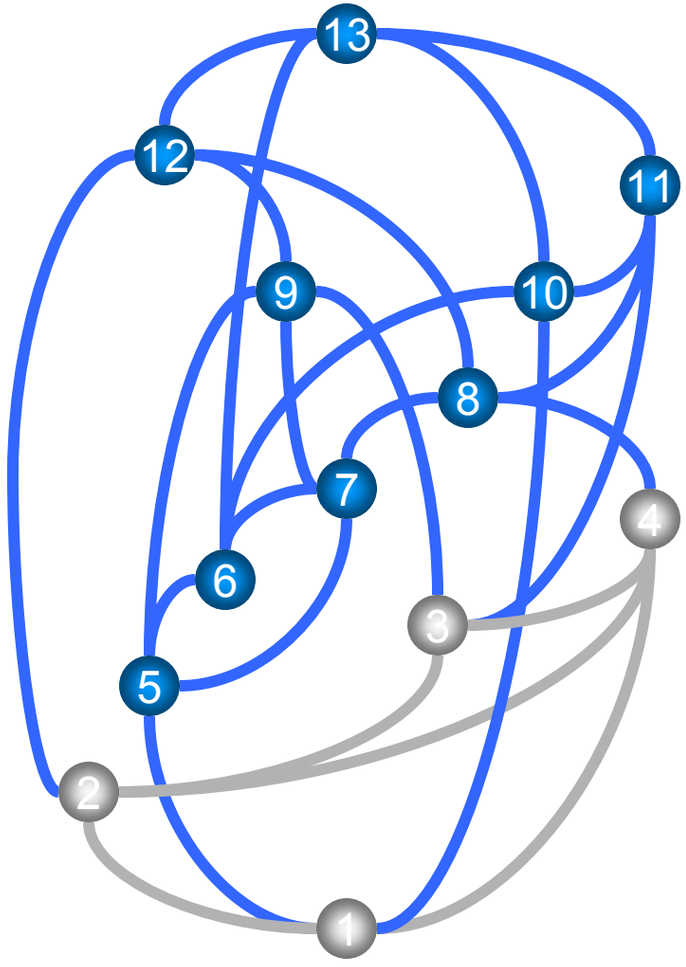


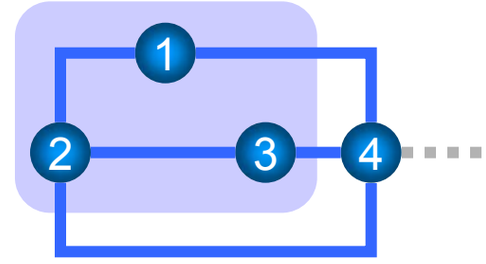


- The first two nodes have a fixed drawing

 copyright©2024 m.patrignani

Example

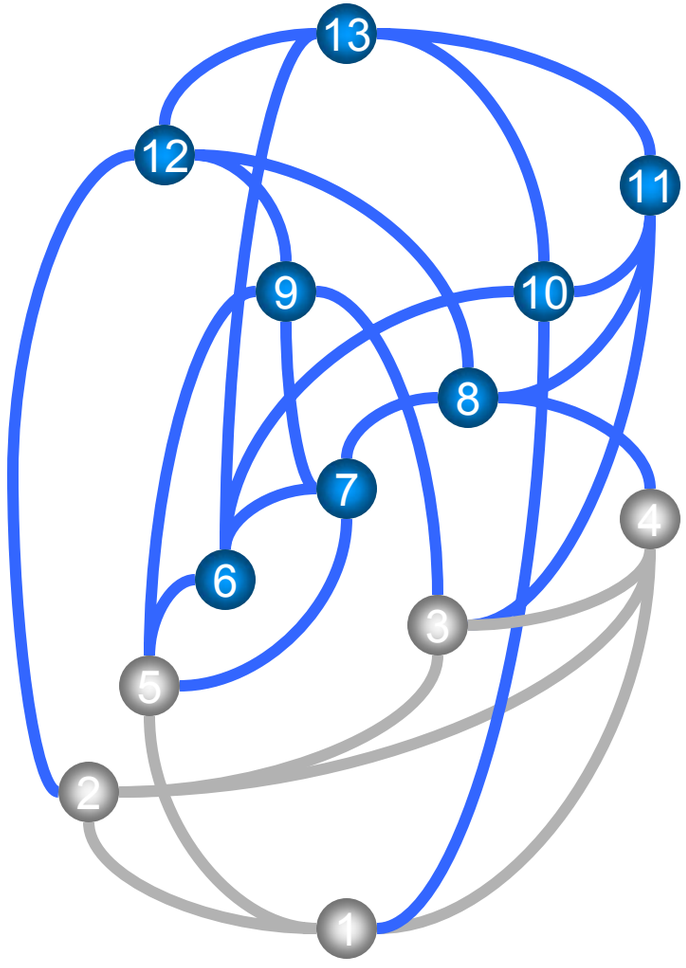


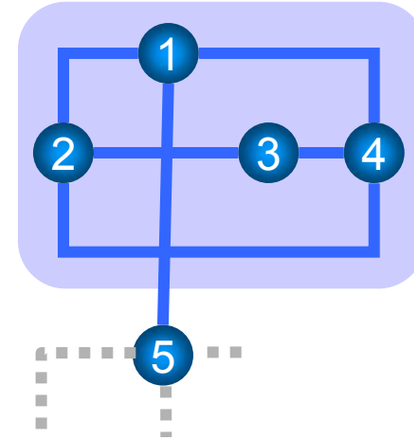


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends

 copyright©2024 m.patrignani

Example

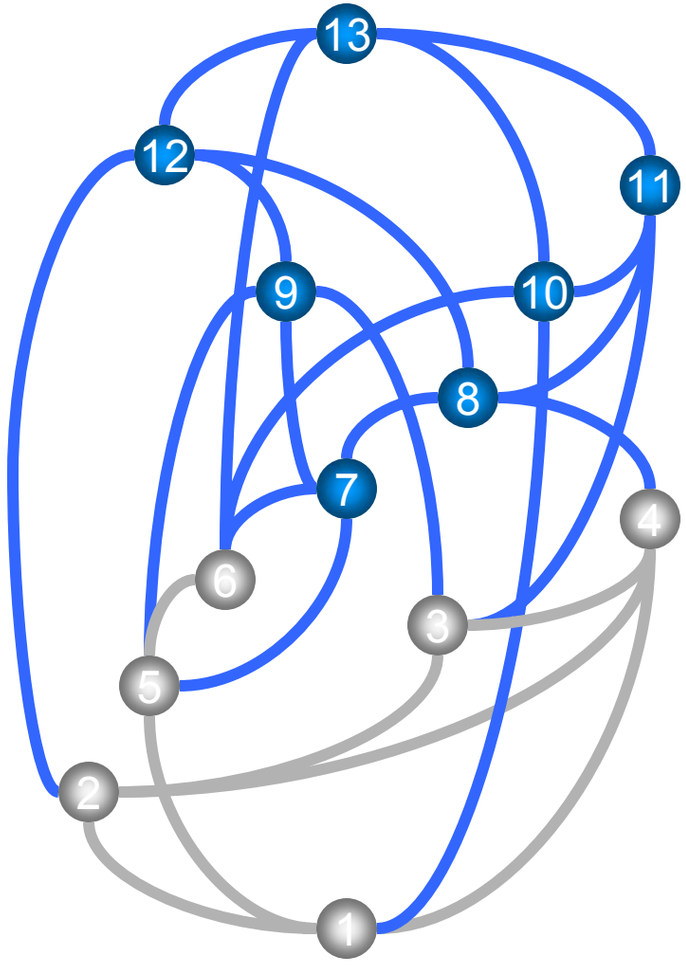


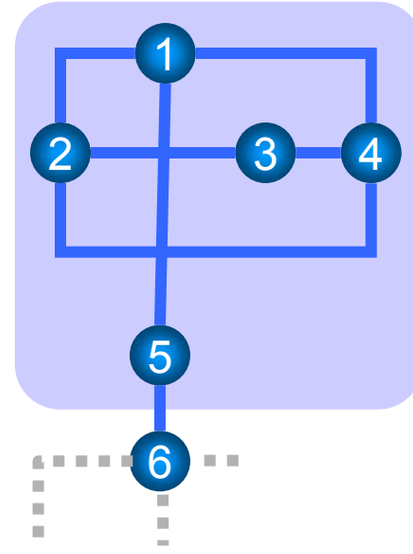


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends

 copyright©2024 m.patrignani

Example

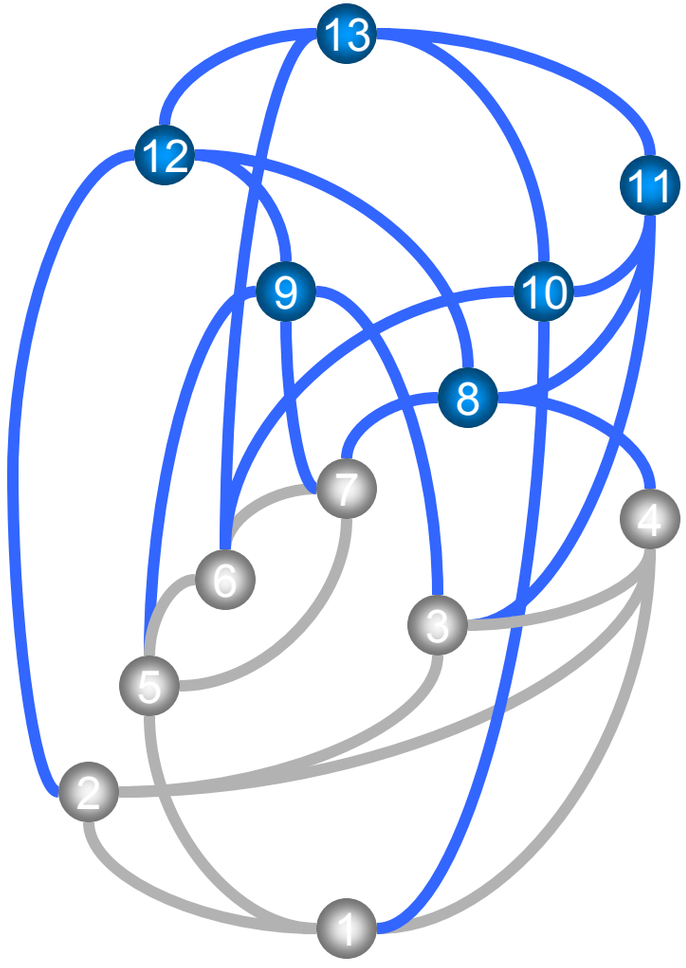


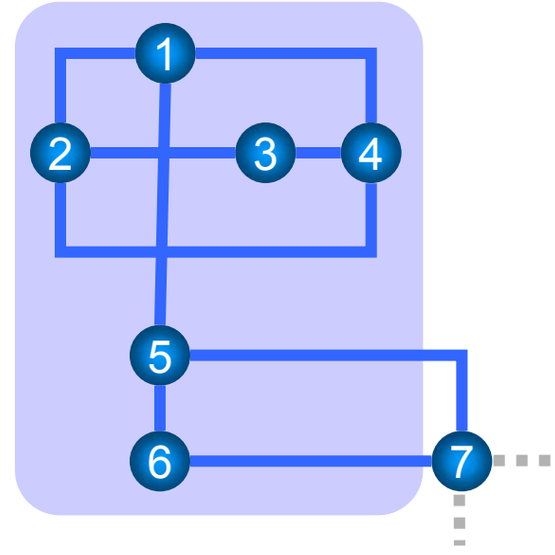


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends

 copyright©2024 m.patrignani

Example

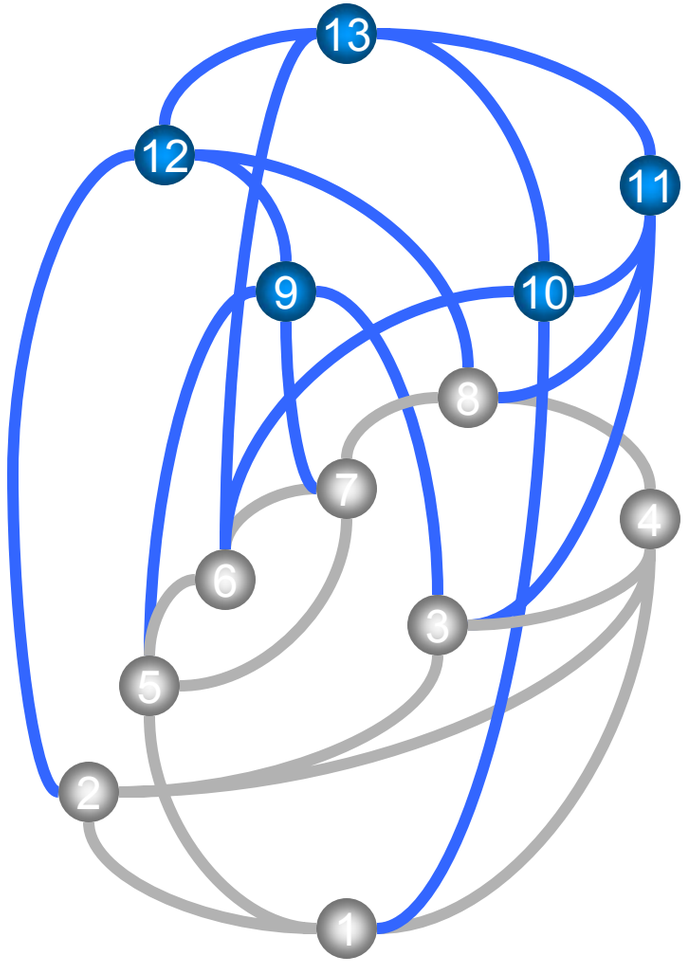


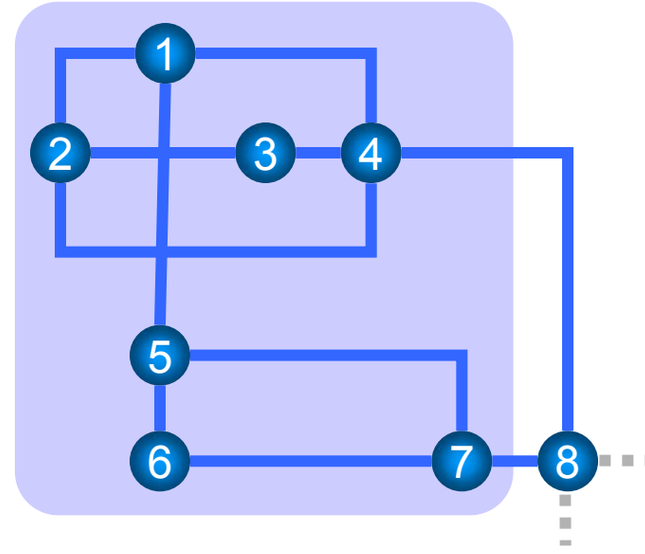


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends

copyright©2024 m.patrignani

Example

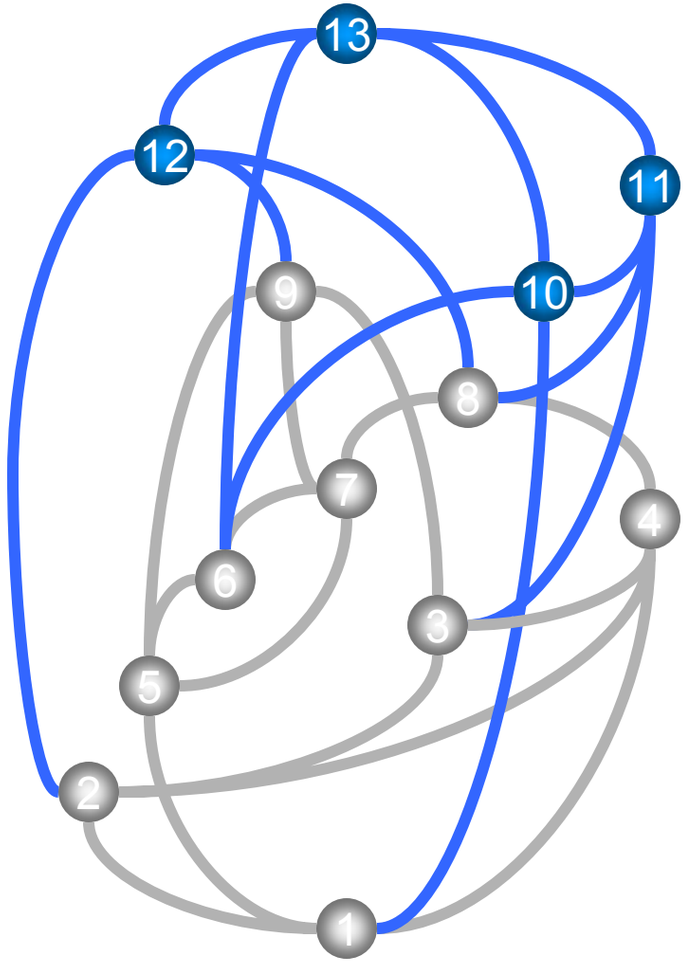


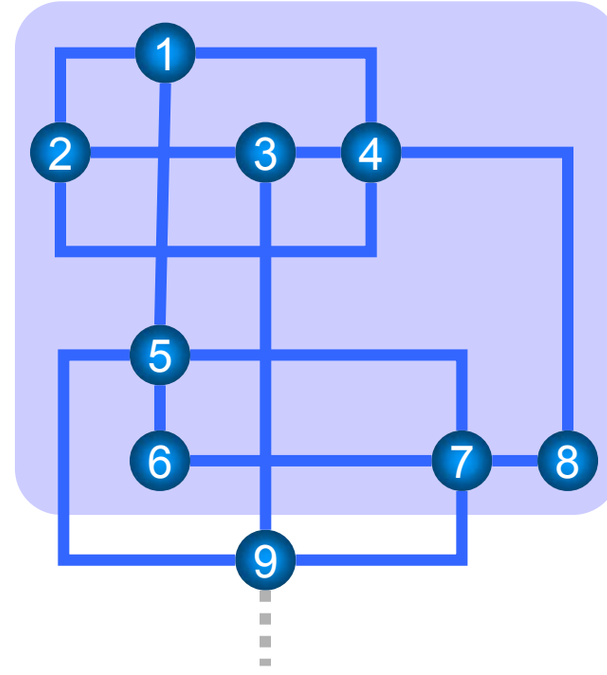


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends

copyright©2024 m.patrignani

Example

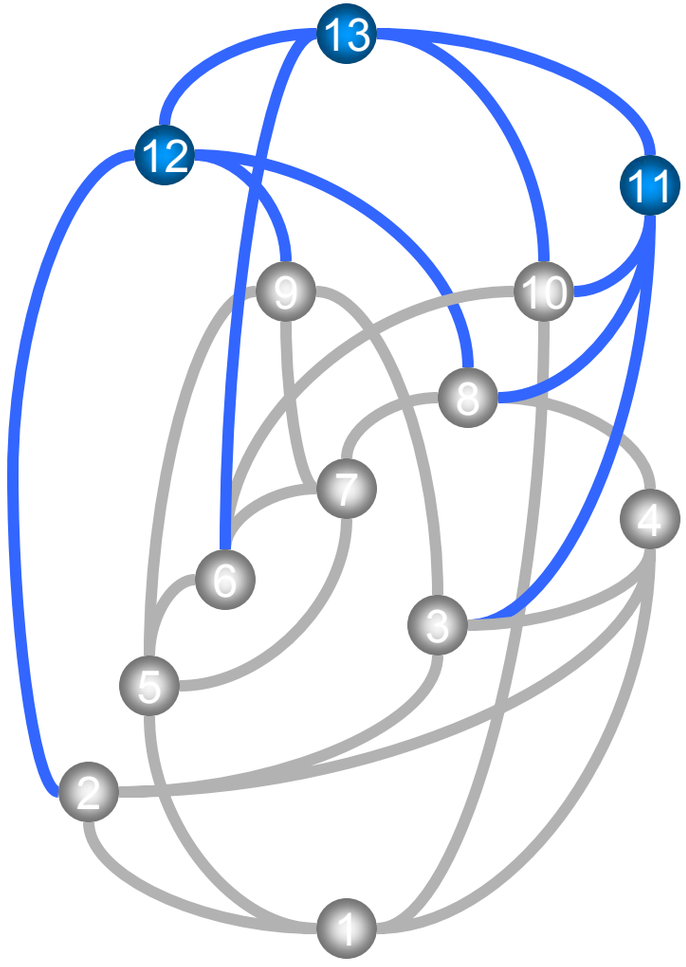


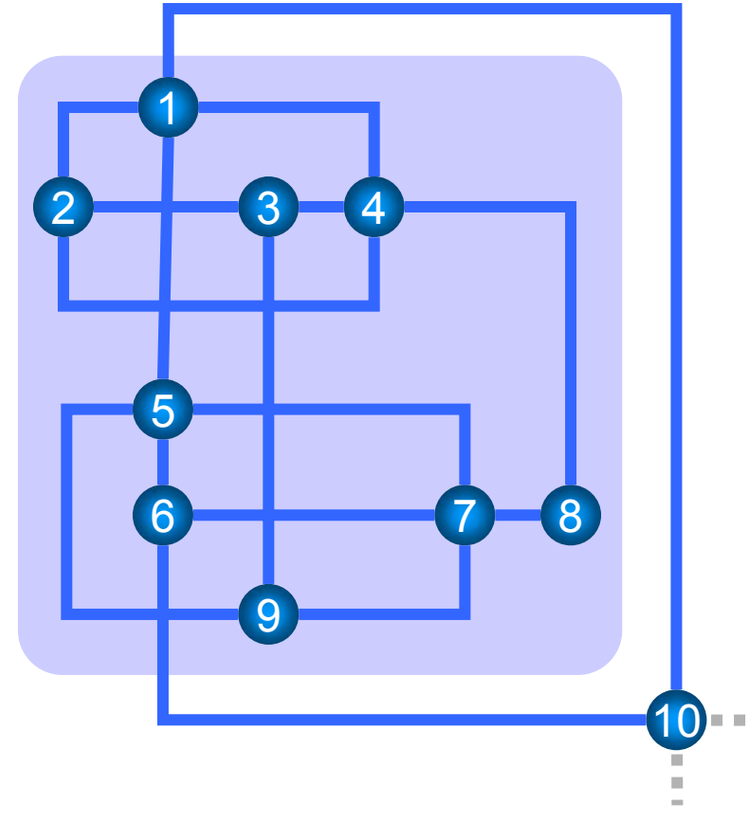


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends

 copyright©2024 m.patrignani

Example

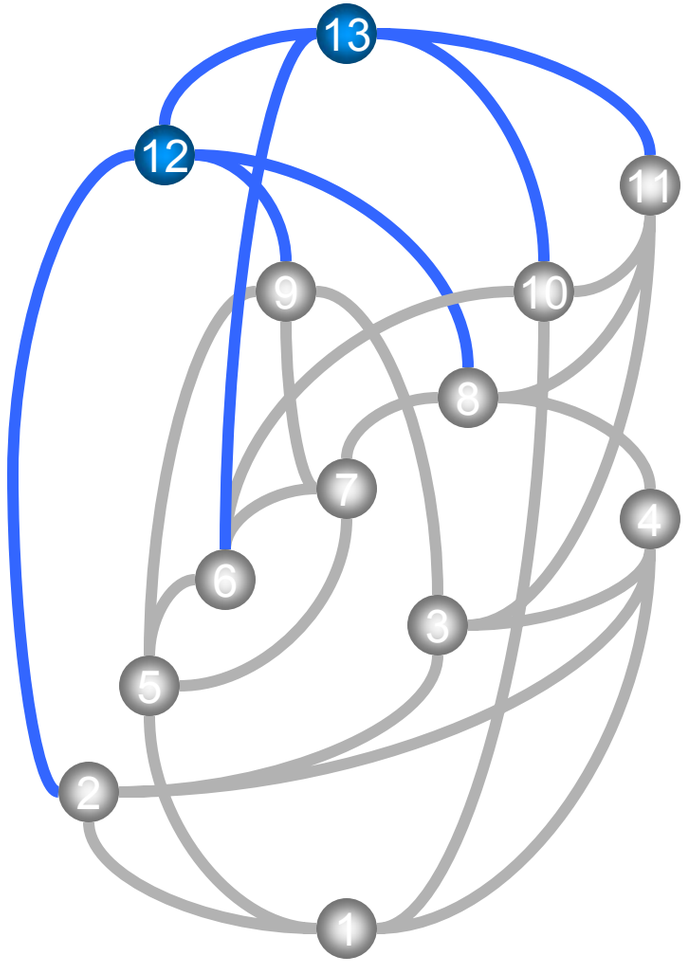


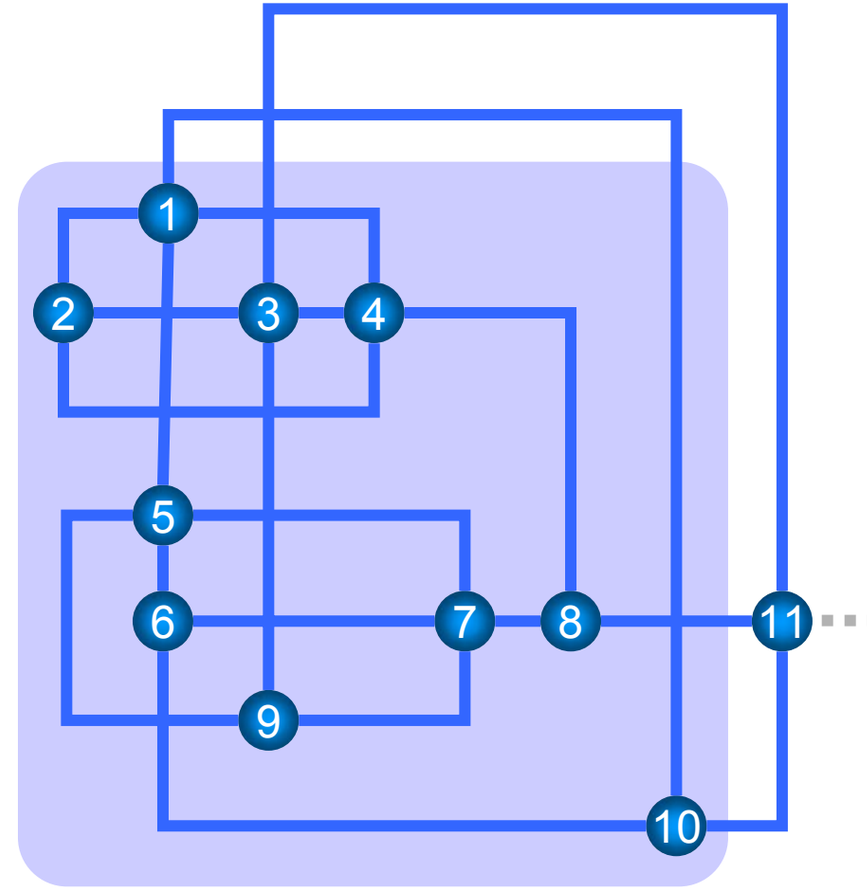


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends

copyright©2024 m.patrignani

Example

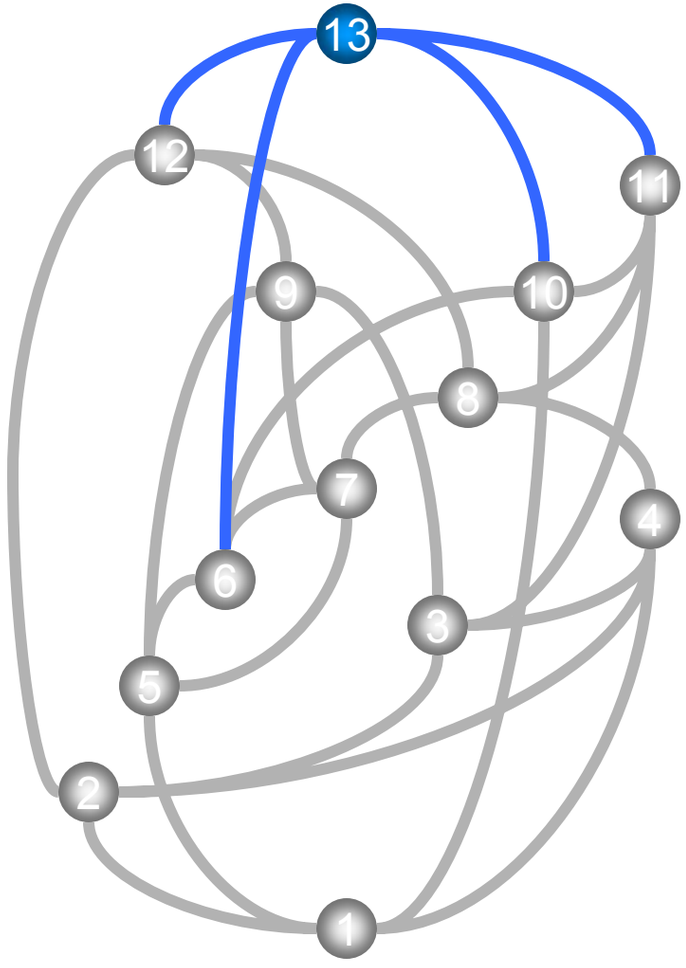


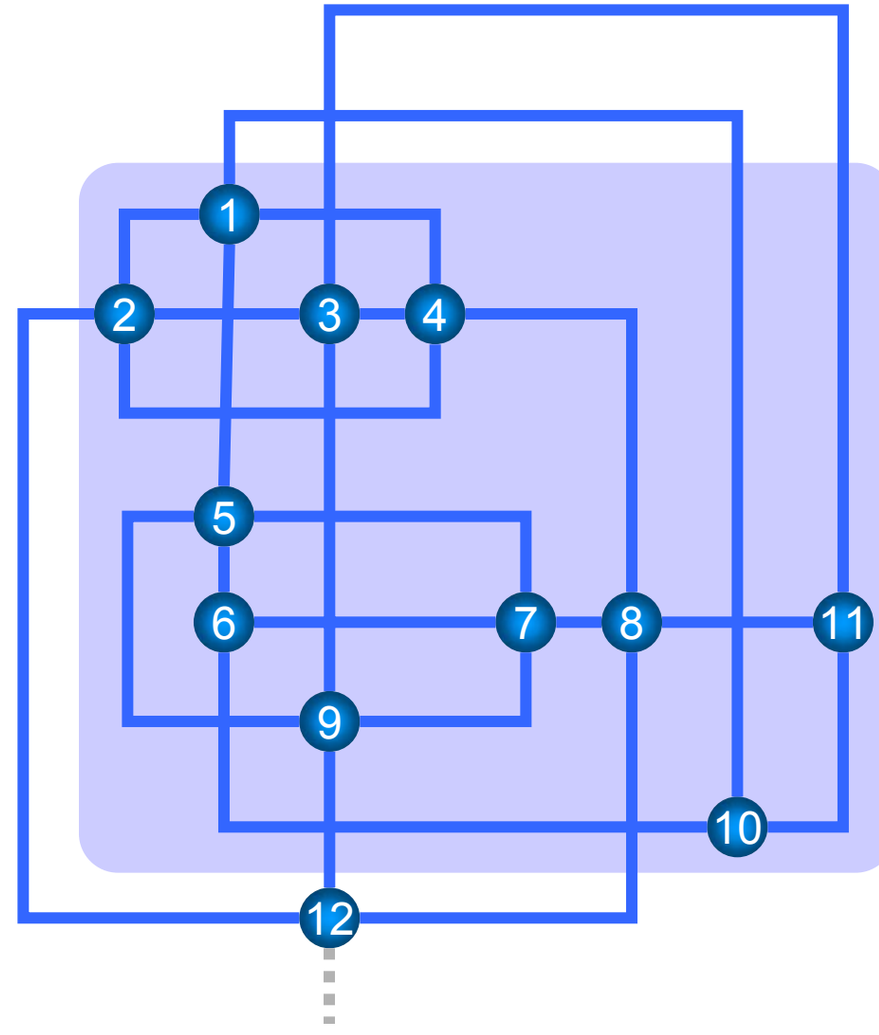


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends

copyright©2024 m.patrignani

Example

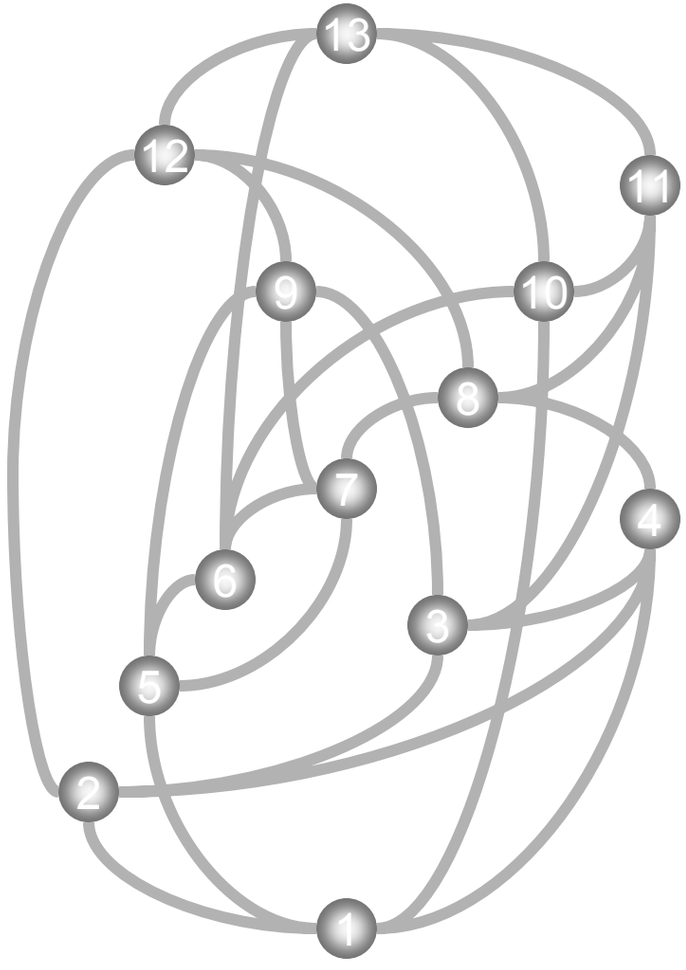


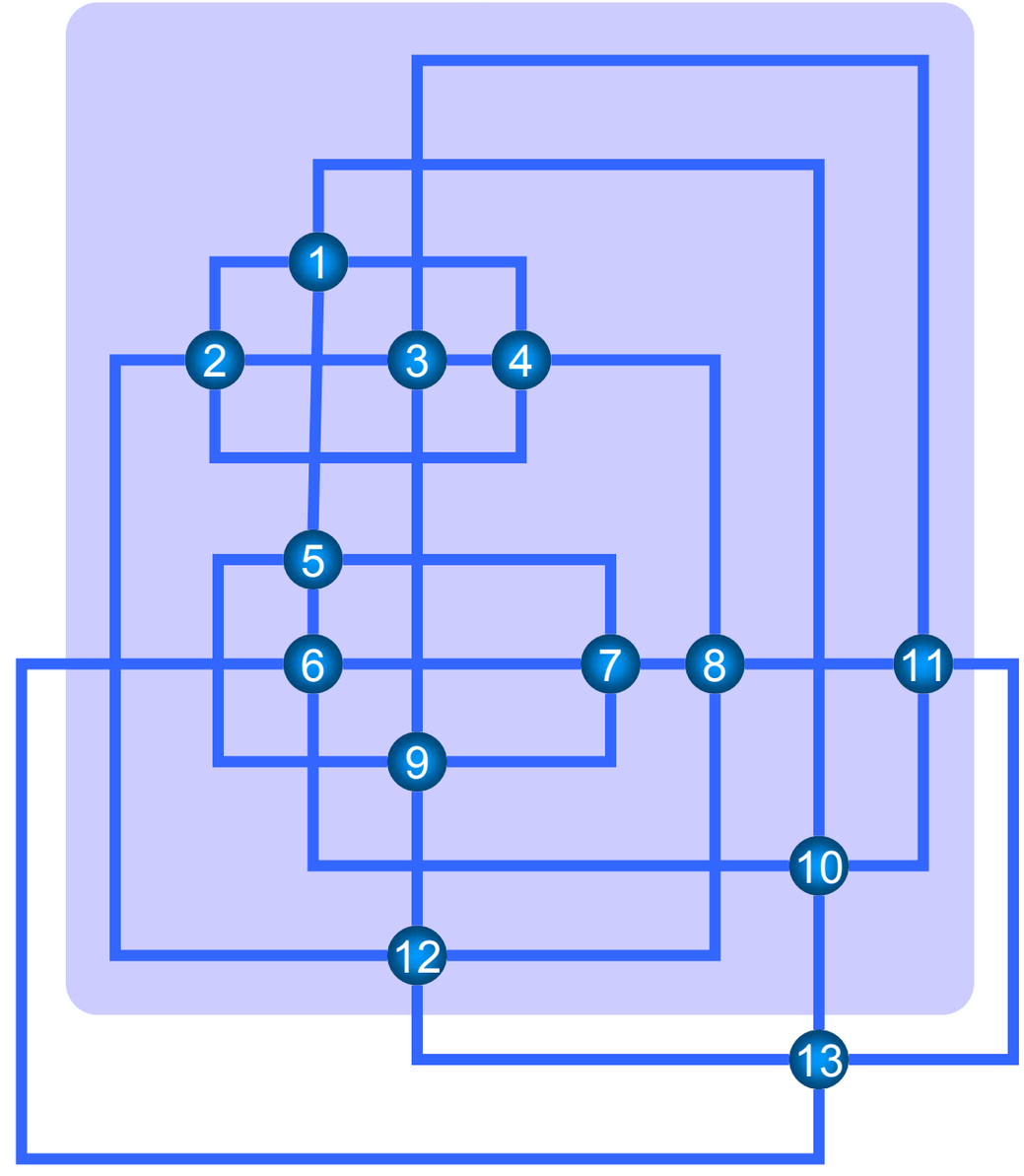


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends

copyright©2024 m.patrignani

Example

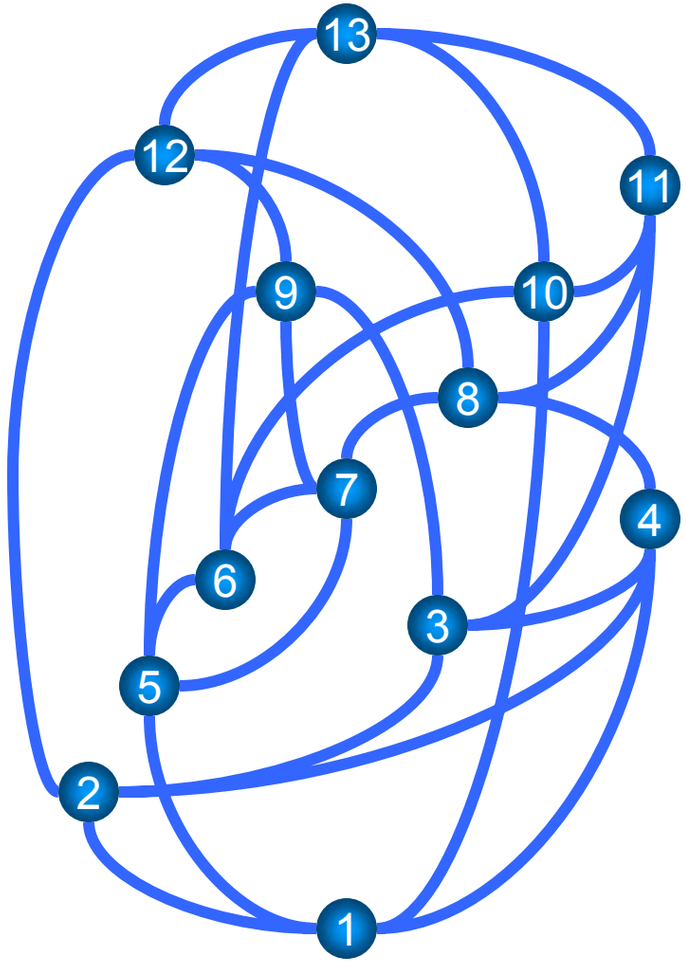


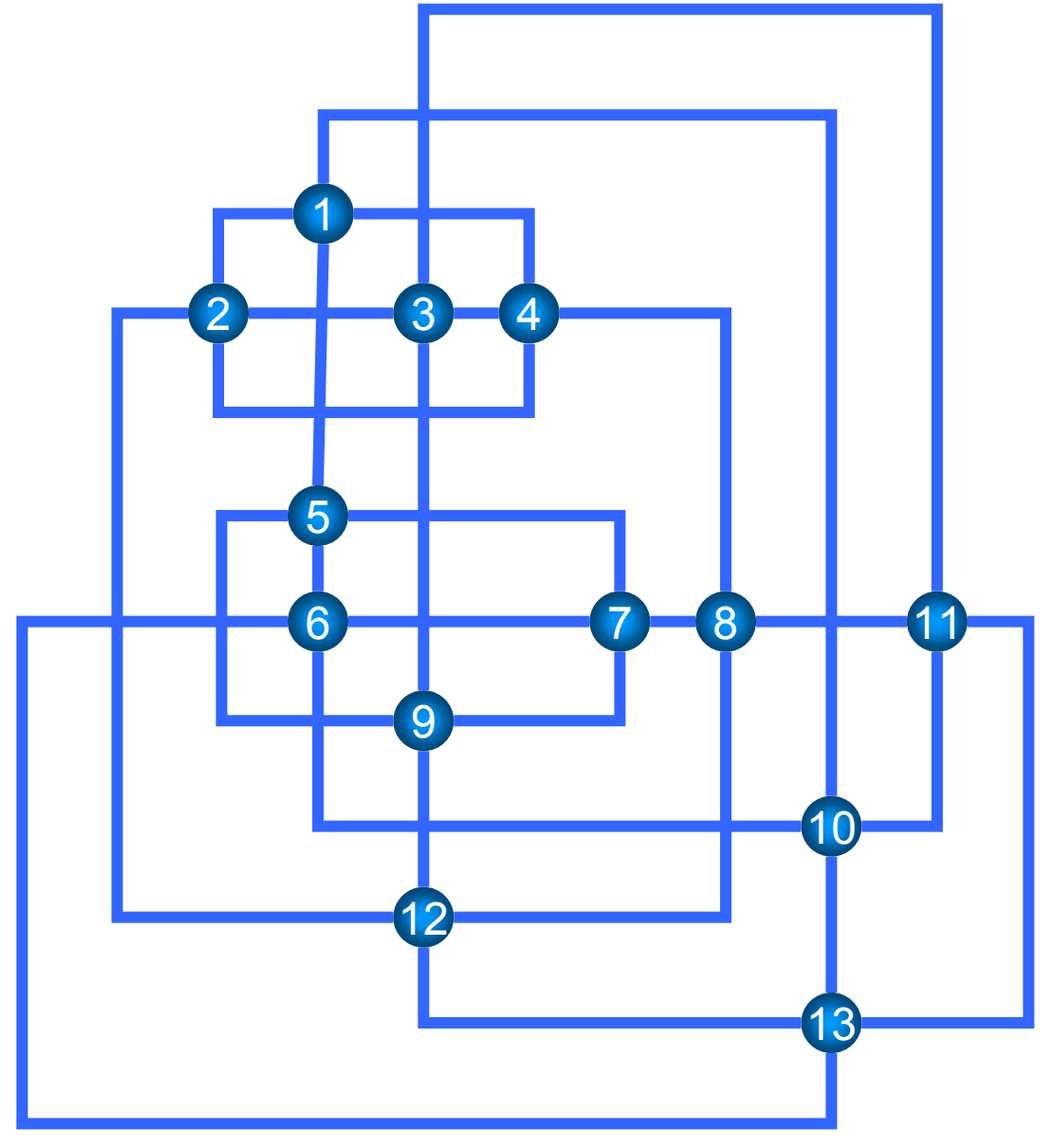


- The first two nodes have a fixed drawing
- Edge (2,4) requires two bends
- Edge (6,13) requires three bends

copyright©2024 m.patrignani

Example





- The final drawing has
  - Width = 10
  - Height = 11
  - Bends = 24

copyright©2024 m.patrignani

Relative-Coordinates Scenario

- Introduced in [Papakostas & Tollis, 96]
- Place new node v in such a way to minimize rows, columns and bends when routing all its edges
  - node v can be placed inside the drawing, making room for it
- Very complex case study with different cases to consider

 copyright©2024 m.patrignani

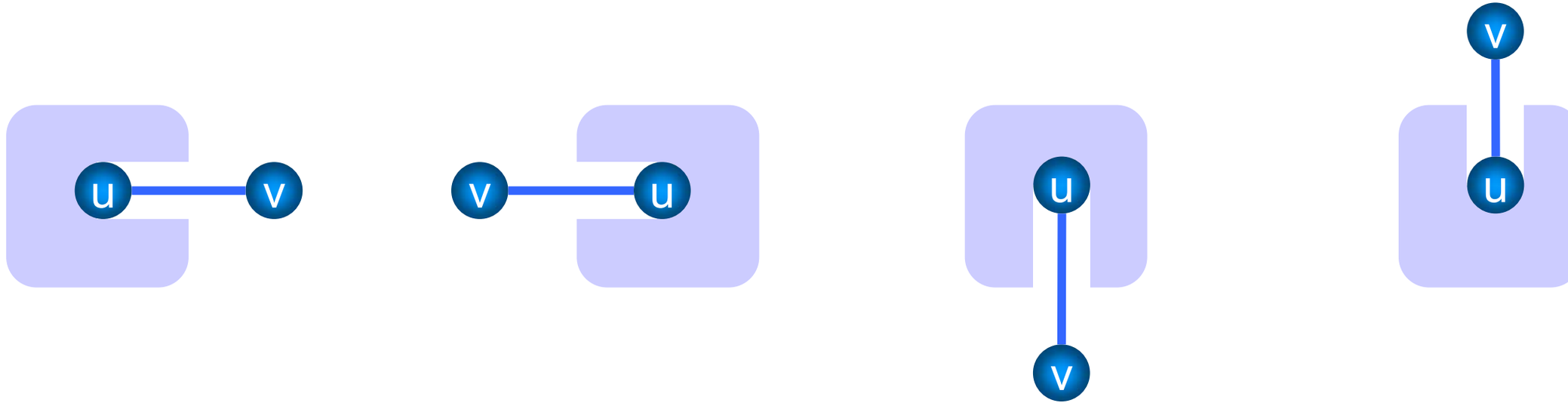
Drawing invariant

- As for the No-Change scenario
  - unused ports of nodes are directly reachable from ouside
- Differently from the No-Change scenario
  - new nodes can be placed:
    - all around the boundary of current drawing
    - inside the current drawing
  - new columns and rows can be inserted inside the current drawing to make room for nodes and bends (preserving the invariant)

 copyright©2024 m.patrignani

Adding a degree-one node

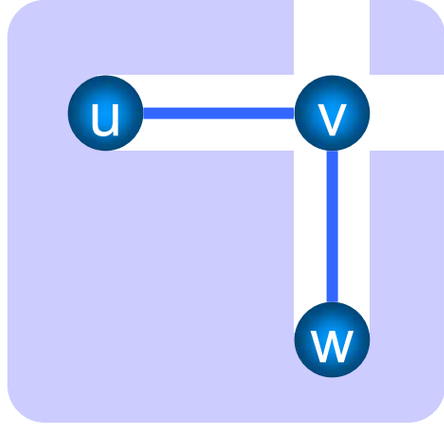
- Add a new row or column outside the current drawing
- Directly connect v to a free port of the adjacent node u

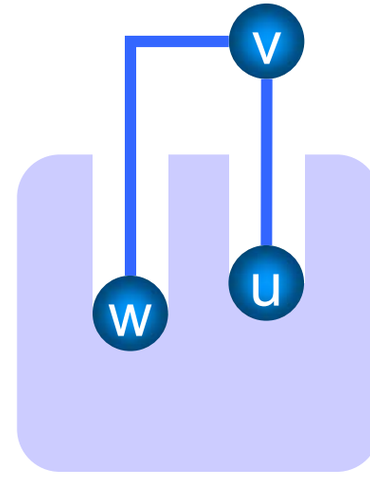


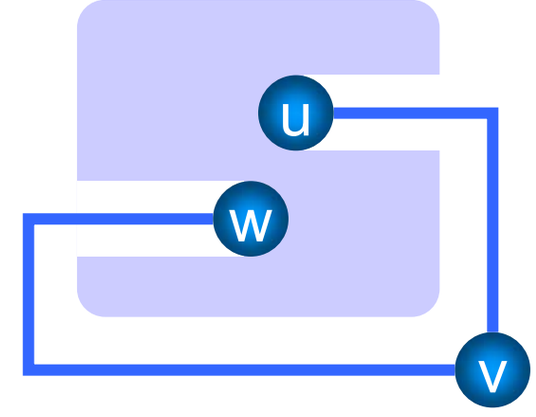
 copyright©2024 m.patrignani

Adding a degree-two node

- In addition to the same configurations of the No-Change scenario:
  - we have cases when no new row or column is needed
  - the new node may be placed outside the bounding on the first top or first right grid line



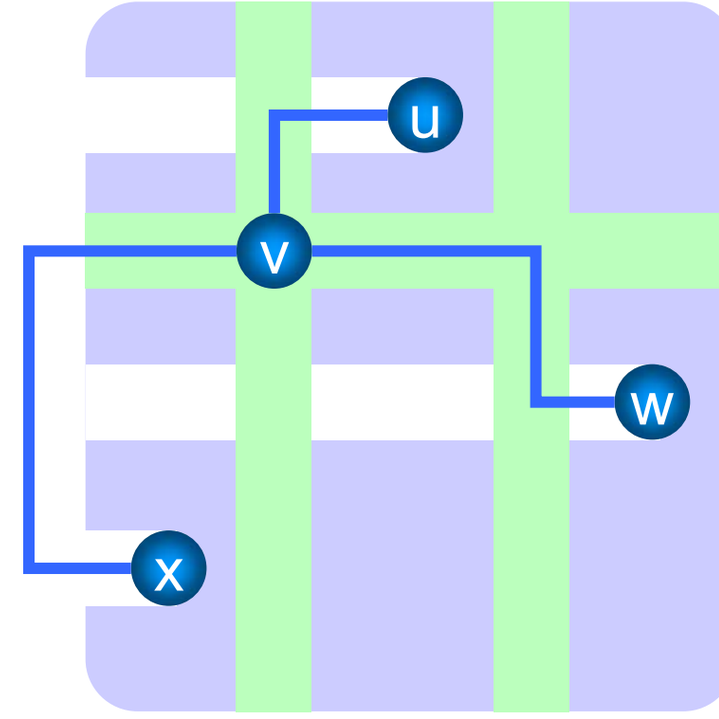




 copyright©2024 m.patrignani

Adding a degree-three node

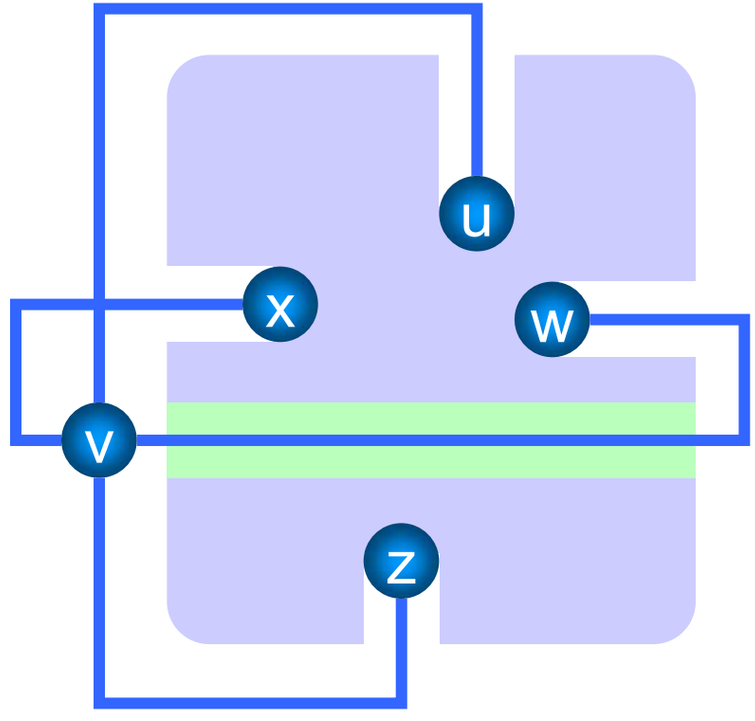
- In the worst case we need
  - four new rows or columns
  - five new bends
    - at most two for the same edge

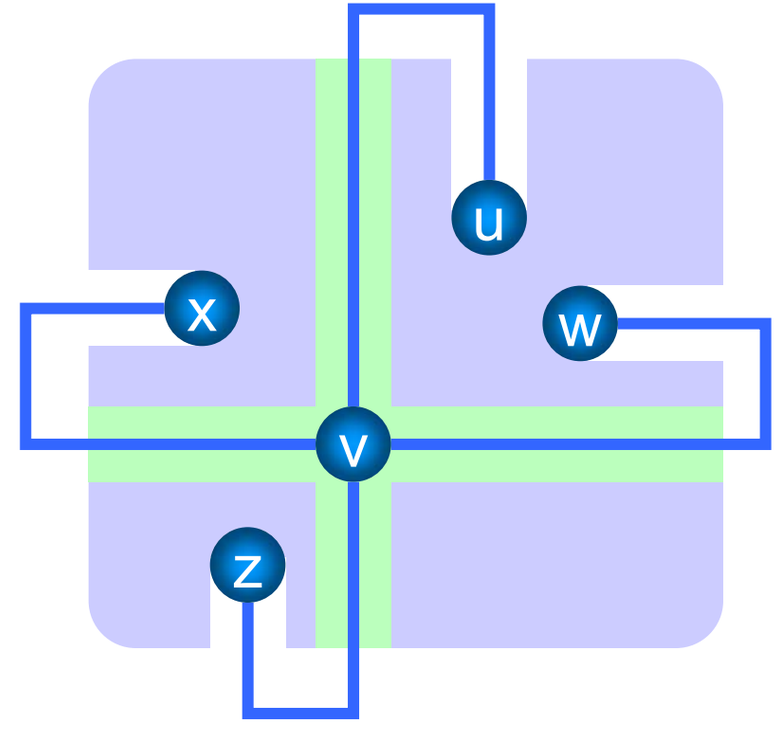


 copyright©2024 m.patrignani

Adding a degree-four node

- In the worst case we need to introduce
  - six new rows and colums (at most four rows)
  - eight new bends

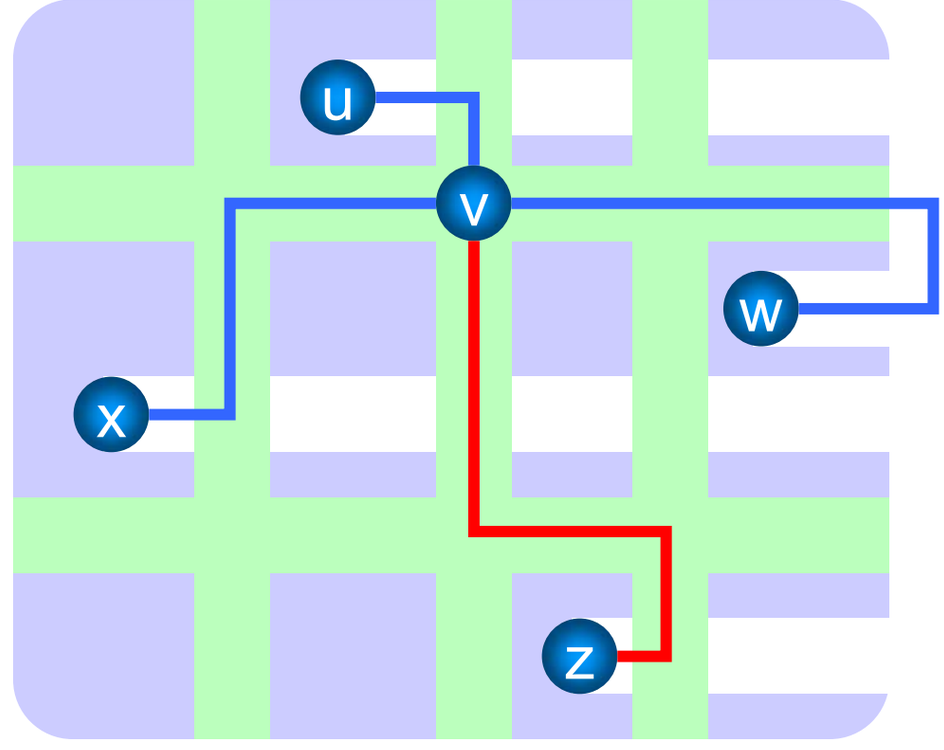




 copyright©2024 m.patrignani

Adding a degree-four node

- In some special case three of the eight new bends are on the same edge



 copyright©2024 m.patrignani

Relative-Coordinates performances

- It can be proved that the Relative-Coordinates drawing method produces drawings with
  - at most 3 bends per edge
  - at most 3n-1 total bends
  - at most $2.25n^2$ area

 copyright©2024 m.patrignani

Example

- Start from an isolated node



 copyright©2024 m.patrignani

Example

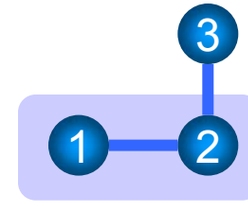
- Start from an isolated node



 copyright©2024 m.patrignani

Example

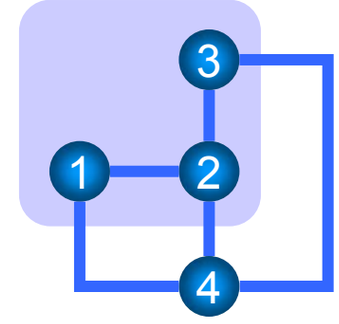
- Start from an isolated node



 copyright©2024 m.patrignani

Example

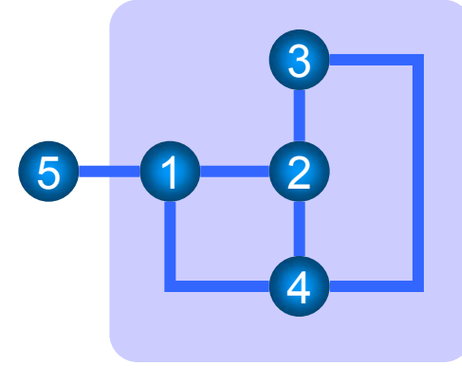
- Start from an isolated node



 copyright©2024 m.patrignani

Example

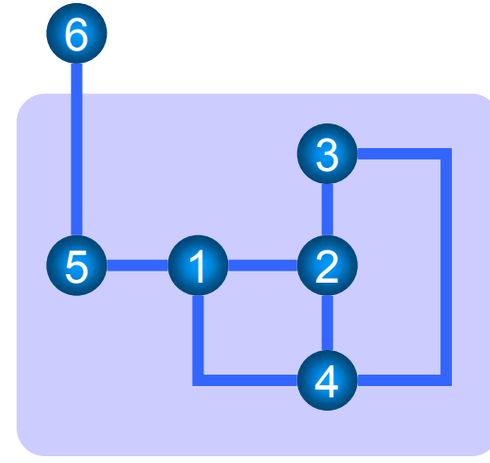
- Start from an isolated node



 copyright©2024 m.patrignani

Example

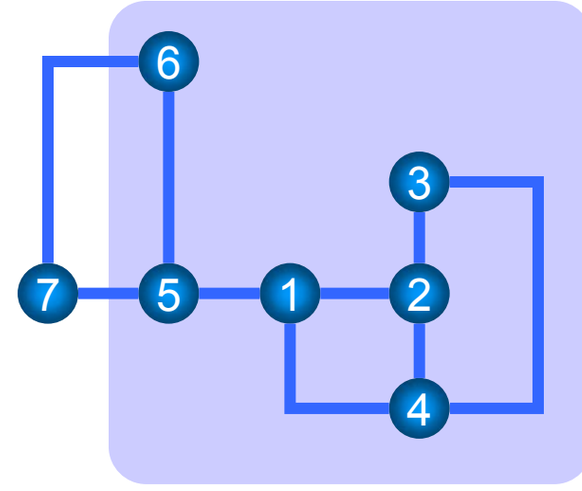
- Start from an isolated node



 copyright©2024 m.patrignani

Example

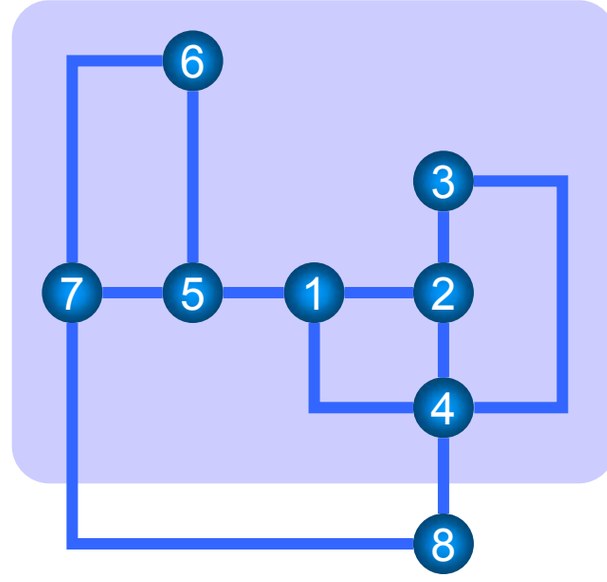
- Start from an isolated node



 copyright©2024 m.patrignani

Example

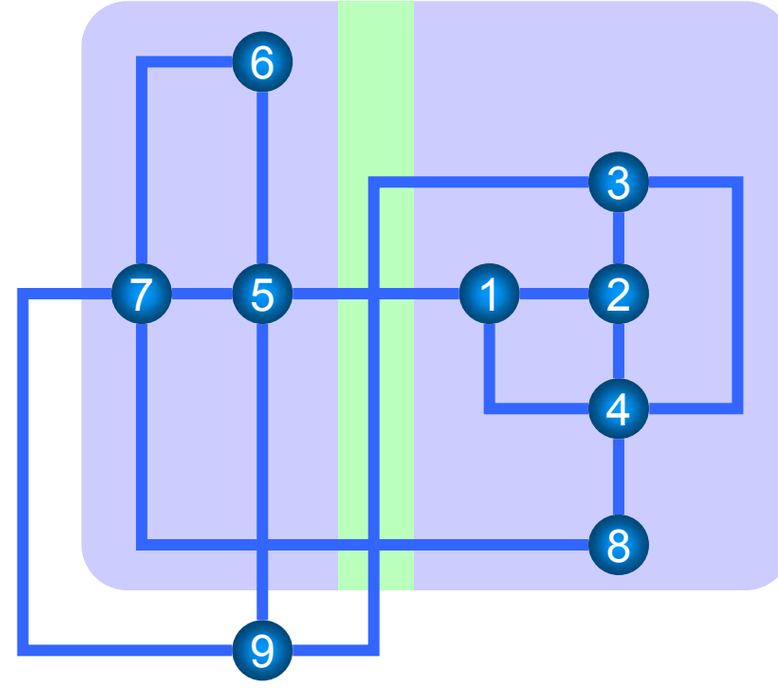
- Start from an isolated node



 copyright©2024 m.patrignani

Example

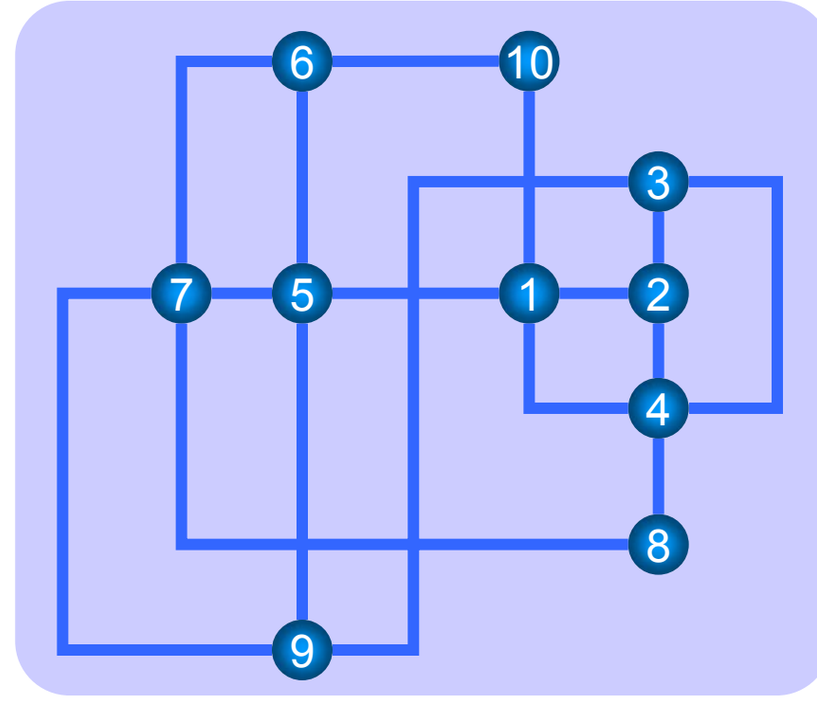
- Start from an isolated node
- Node 9 requires a new column to be inserted inside the drawing



 copyright©2024 m.patrignani

Example

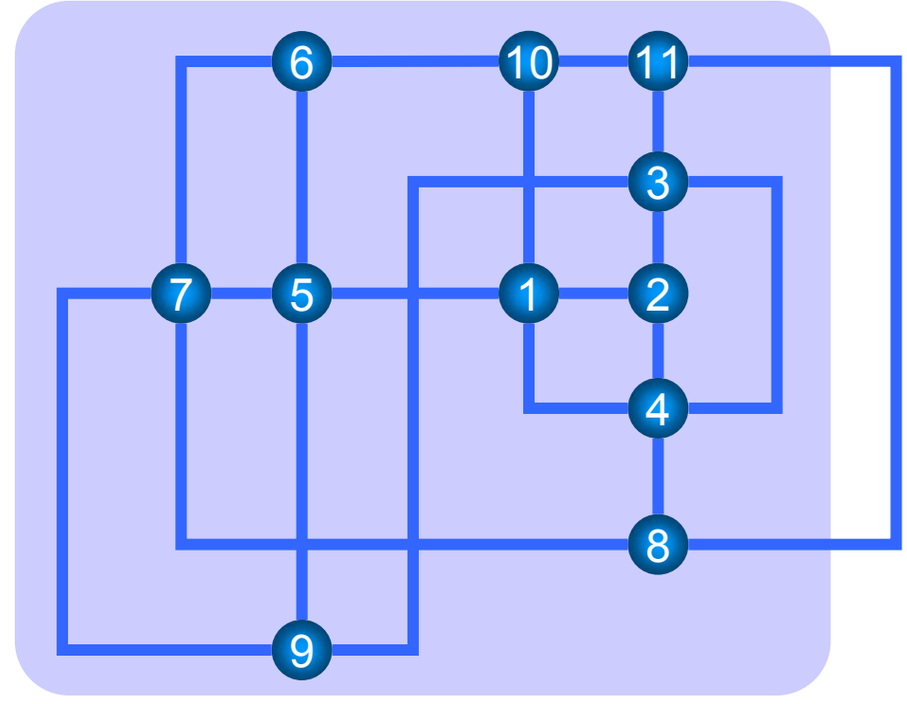
- Start from an isolated node
- Node 9 requires a new column to be inserted inside the drawing



 copyright©2024 m.patrignani

Example

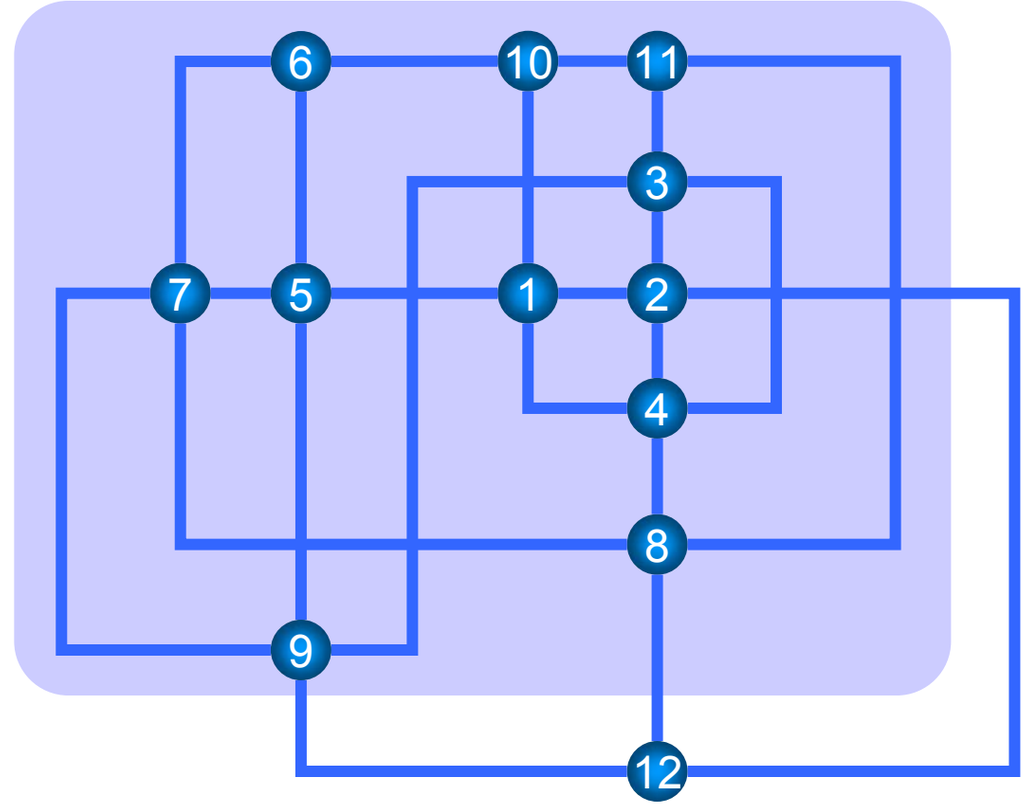
- Start from an isolated node
- Node 9 requires a new column to be inserted inside the drawing



 copyright©2024 m.patrignani

Example

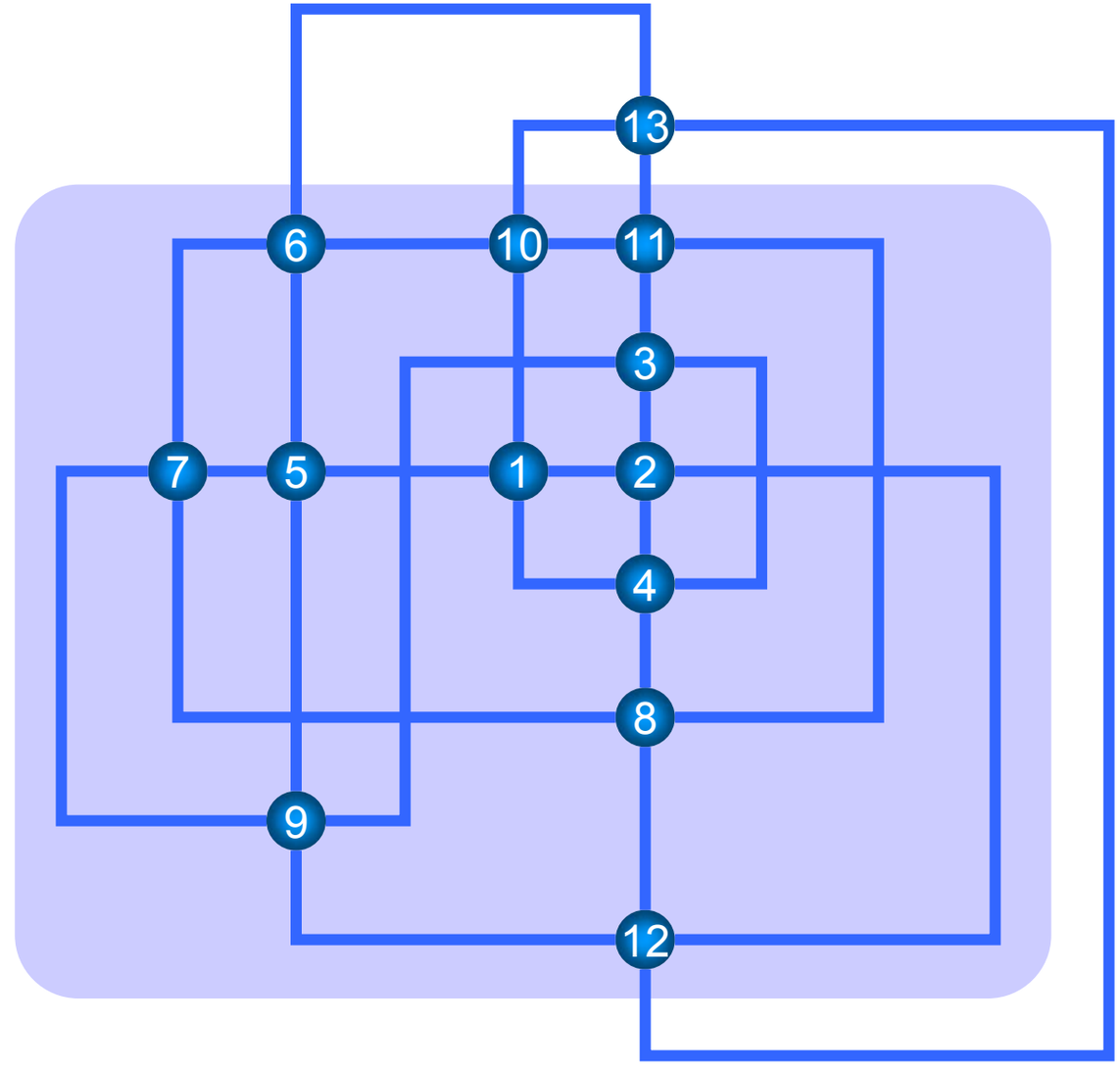
- Start from an isolated node
- Node 9 requires a new column to be inserted inside the drawing



 copyright©2024 m.patrignani

Example

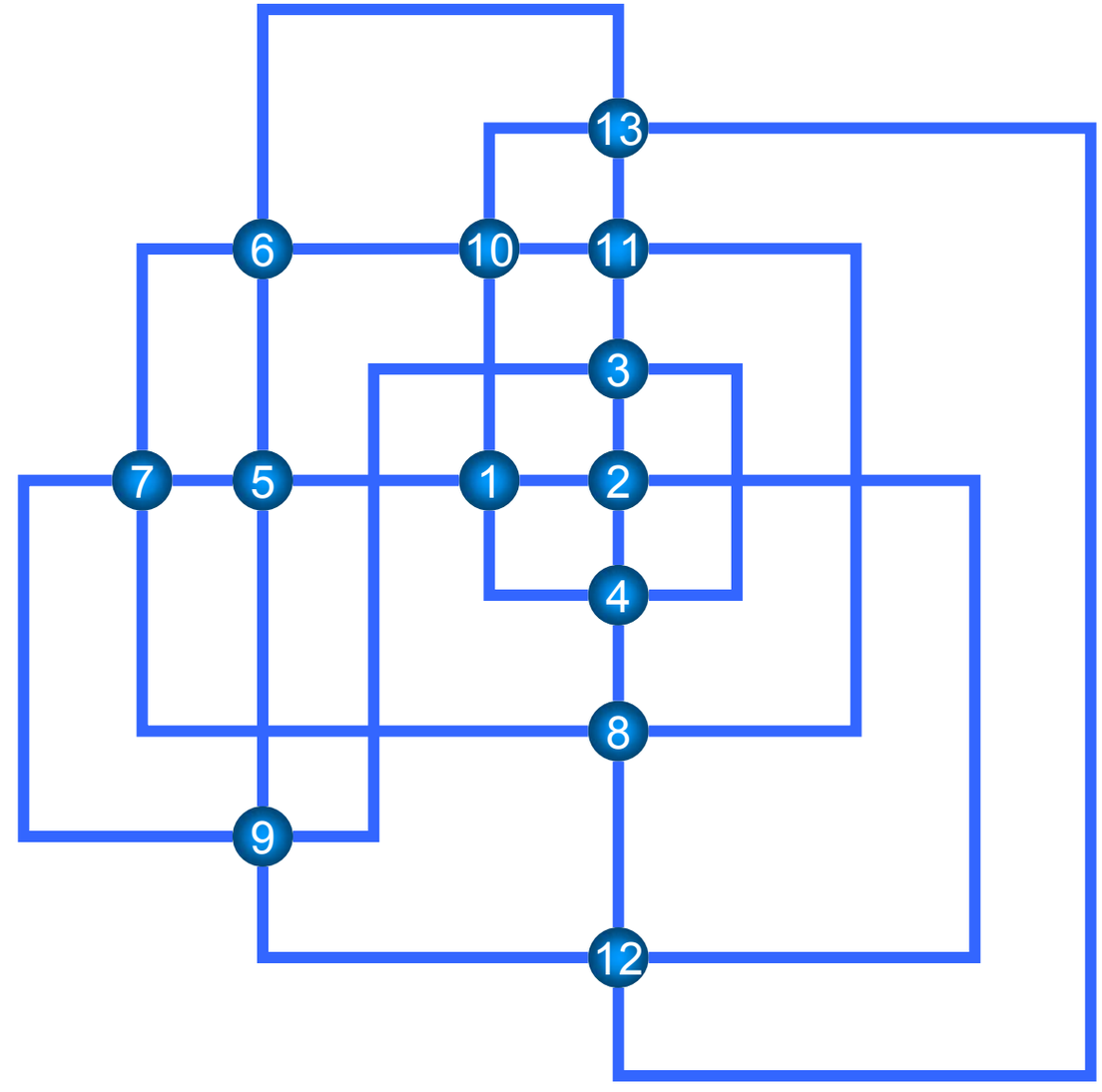
- Start from an isolated node
- Node 9 requires a new column to be inserted inside the drawing
- Edge (12,13) requires three bends



 copyright©2024 m.patrignani

Example

- The final drawing has
  - Width = 9
  - Height = 9
  - Bends = 20



 copyright©2024 m.patrignani

Algorithm Four-Orthogonal

- More sophisticated techniques (that require the graph to be biconnected) guarantee [Di Battista *et al.*, '99]
  - running time O(n)
  - area $O((7n+10)^2 / 64) \sim O(0.77n^2)$
  - at most 2n+4 bends in total
  - at most 2 bends per edge

 copyright©2024 m.patrignani

Bibliography and further readings

- [Biedl & Kant, 94] Biedl, T. C. and Kant, G. “A better heuristic for orthogonal graph drawings”%. ESA 1994.
- [Di Battista *et al.*, ‘99] G. Di Battista, P. Eades, R. Tamassia, and I. G. Tollis, “Graph Drawing: Algorithms for the Visualization of Graphs”, Prentice-Hall, 1999
- [Even & Tarjan, ‘76] S. Even and R. E. Tarjan. Computing an st-numbering. Theoret. Comput. Sci., 2, 1976
- [Fialko & Mutzel, 98] S. Fialko and P. Mutzel. A new approximation algorithm for the planar augmentation problem. SODA 1998
- [Garg & Tamassia, ‘96] A. Garg and R. Tamassia, “A new minimum cost flow algorithm with applications to graph drawing”, Graph Drawing, 1996
- [Lempel *et al.*, ‘67] A. Lempel, S. Even, and I. Cederbaum, “An algorithm for planarity testing of graphs”, Theory of Graphs: International Symposium 1967

 copyright©2024 m.patrignani

Bibliography and further readings

- [Papakostas & Tollis, 96] A. Papakostas and I. G. Tollis, Issues in Interactive Orthogonal Graph Drawing, GD'95, 1996
- [Papakostas & Tollis, 98] A. Papakostas, I. G. Tollis, "Interactive Orthogonal Graph Drawing", *IEEE Trans. Computers*, 47, 1998
- [Papakostas *et al.*, 97] A. Papakostas, J. M. Six, I. G. Tollis, "Experimental and Theoretical Results in Interactive Orthogonal Graph Drawing", Graph Drawing '96, 1997
- [Patrignani 04] Maurizio Patrignani. "A Note on the Self-Similarity of Some Orthogonal Drawings". Graph Drawing 2004
- [Tamassia & Tollis, 86] Roberto Tamassia and Ioannis G. Tollis, "A unified approach to visibility representations of planar graphs", Discrete Comput. Geom., 1, 1986
- [Tamassia & Tollis, 89] Roberto Tamassia and Ioannis G. Tollis, "Planar grid embedding in linear time", IEEE Trans. Circuits Syst., CAS-36, 1989
- [Tamassia 87] Roberto Tamassia "On Embedding a Graph in the Grid with the Minimum Number of Bends", SIAM J. Computing, 16, 1987

 copyright©2024 m.patrignani